# SELD: Detecting Silent Evidence Loss in Context Compression beyond the Answer

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## Abstract

Context compression is now routine for language models with long inputs, but a compressed context is useful only if it keeps the evidence for the answer. Existing checks compare the reader model's answer or its most likely words, and thus miss a frequent failure, *silent evidence loss*: key evidence is removed while the answer stays the same, leaving the answer unsupported and risky for other models that reuse the context. To address this challenge, we propose SELD (Silent Evidence Loss Detector), which needs no training and no labels. Its principle is that when evidence is lost, a reader model's preferences among the answers it leaves unsaid shift even if its answer does not, so comparing these preferences before and after compression reveals losses that answer checks cannot see. We also build SEL-Pairs, a benchmark of length-matched evidence and control deletions from three question-answering datasets. Quantitative and qualitative experiments with five reader models on it show that SELD consistently outperforms strong and recent baselines across compressors, compression rates and models, identifying the evidence-removing edit in 83% of matched pairs with unchanged answers, against 70% for the strongest baseline. These results show that evidence loss in compressed contexts can be detected even when the answer is unchanged.

## 1 Introduction

Language models rarely read the full original text anymore. Retrieved passages are pruned, prompts are shortened by learned token classifiers, and agents pass condensed notes and memories to each other (Jiang et al., 2023b; Pan et al., 2024; Xu et al., 2024; Chirkova et al., 2025). Each step can drop the evidence that a question depends on, so a deployed system needs to know, without gold answers, whether a compressed context still supports the answer. The common safeguard is to ask the model again: if the answer with the compressed context matches the answer with the full context, the compression is assumed to be harmless (Manakul et al., 2023; Cohen-Wang et al., 2024).

This safeguard has a blind spot (Figure 1). Suppose a US travel guide says *Drive on the right* and the question is *Which side of the road do cars drive on?* After compression removes this sentence, reader model A still answers *the right* from its default, so the answer check passes. The compressed context is then passed to reader model B, which also answers *the right* with the full guide, but now falls back on a different default and answers *the left*. The answer of A survived; its evidence did not. We call this *silent evidence loss* (SEL). It is common: deleting a key sentence and, separately, an equally long non-key sentence leaves the answer unchanged both times in 63–78% of pairs in our development data, and on MuSiQue (Trivedi et al., 2022) about one in five correct answers reappears with an officially unanswerable context. Such downstream effects are measurable: a key-sentence deletion that leaves the answer of one model unchanged changes the answers of *other* models more often than a matched non-key deletion in 6 of 10 cross-family pairs in our data.

That answers can survive missing evidence is known from multi-hop question answering (QA) and retrieval-augmented generation (Trivedi et al., 2020; Lee et al., 2021; Joren et al., 2025), and recent work argues that final answers hide the effect of evidence (Jain and Vedam, 2026). What is missing is a label-free way to *see* SEL when a context is compressed. Checks based on the answer, such as answer agreement or the likelihood of the original answer (Cohen-Wang et al., 2024), are blind to SEL by construction: they observe the reader model only through the answer that did not change. Scores that compare output distributions, such as Jensen–Shannon (JS) divergence (Lin, 1991) or the recent LUMINA (Yeh et al., 2026), weight tokens

by probability, so they are dominated by the few most likely tokens and inherit much of the same blindness. Evaluation is a further obstacle: existing benchmarks do not separate evidence loss from mere text loss, so a detector can look good simply because removing more text changes more.

An unchanged answer only means that the same candidate still wins, much like an election in which the winner stays while the vote counts shift. We therefore read the reader model's belief over all candidates and ask whether its pattern, not its winner, has changed. Our contributions are as follows.

- **Problem and benchmark.** We identify and define *silent evidence loss* (Definition 1), a frequent failure of context compression in which key evidence is removed but the answer stays the same. To measure it, we build **SEL-Pairs**, a benchmark from three public multi-hop QA datasets (HotpotQA, 2WikiMultihopQA and MuSiQue) in which every evidence deletion is matched by an exactly equally long control deletion from the same context, both for single sentences and inside real outputs of two compressors, so that a detector cannot win simply because more text was removed. On it, answer checks miss most silent evidence loss (Section 5.1).
- **Method.** We show that output divergences mix a change in the reader model's confidence with a change in the *pattern* of its preferences (Proposition 1), and find that lost evidence shows in the pattern. SELD (Silent Evidence Loss Detector) isolates this pattern; it needs no training, labels or generated answer, and one forward pass (Section 3).
- **Experiments.** On SEL-Pairs with five reader models, SELD beats JS divergence by 12.4 points and the answer-likelihood drop by 28.8 points in a test fixed before data collection (Section 5.2), and beats JS divergence and an adapted LUMINA detector in every fixed test on real compressor outputs that could be estimated (Section 5.3); simpler explanations do not account for the gains (Section 5.4).
- **Further findings.** Lost evidence leaves its trace in the answers the reader model would almost never say: the signal survives removing the 1,000 most likely tokens and reappears in the reader model's hidden state (Section 5.5). SELD and answer agreement are complementary, and combined they flag lost and silently lost evidence better than either alone (Section 5.6).

## 2 Related Work

**Compressing contexts and checking the result.** Compressors range from token pruning (Li et al., 2023; Jiang et al., 2023b, 2024; Pan et al., 2024) and question-aware sentence pruning (Chirkova et al., 2025) to learned and task-aware methods (Xu et al., 2024; Chuang et al., 2024; Liskavets et al., 2025; Xu et al., 2025; Tang et al., 2026) and key–value (KV) cache eviction (Li et al., 2024; Kim et al., 2025); they are evaluated mainly by answer accuracy, and studies ask what information they keep (Łajewska et al., 2025). Text-side frameworks verify that units of meaning survive (Trukhina and Vashkelis, 2026), QAGS, FEQA and QuestEval check summaries with generated questions (Wang et al., 2020; Durmus et al., 2020; Scialom et al., 2021), and trained checkers read representations (Zeng et al., 2025) or teach a model to abstain when KV compression evicts evidence (Khodabandehlou and Krishnamachari, 2026). In multi-step agents, concurrent work finds that compaction silently drops constraints and state, detected by task outcomes (Chen, 2026), LLM judges (Wang et al., 2026; Chen et al., 2026) or paired environment replays (Min et al., 2026). SELD needs no labels, training or replay and could serve as their feature.

**Answers without evidence.** Multi-hop QA models answer through disconnected reasoning (Trivedi et al., 2020), reader models can be right without evidence (Lee et al., 2021), and language models are often correct with insufficient context (Joren et al., 2025). CUE-R removes and duplicates single evidence items and argues that the final answer hides their effect (Jain and Vedam, 2026), without controlling prompt length or comparing output distributions.

**Comparing output distributions.** Context attribution scores the likelihood of a fixed answer under ablated contexts (Cohen-Wang et al., 2024). PECoRe and Du et al. compare predictions with and without context by Kullback–Leibler (KL) divergence or mutual information (Sarti et al., 2024; Du et al., 2024); LUMINA measures context use through an embedding-based distance over the most likely tokens (Yeh et al., 2026); distribution-based perturbation tests compare sampled answers (Rauba et al., 2025); and for quantized models, divergences of the full next-token distribution reveal changes that accuracy hides (Qamar et al., 2026). The cosine of first-token logits has served as an

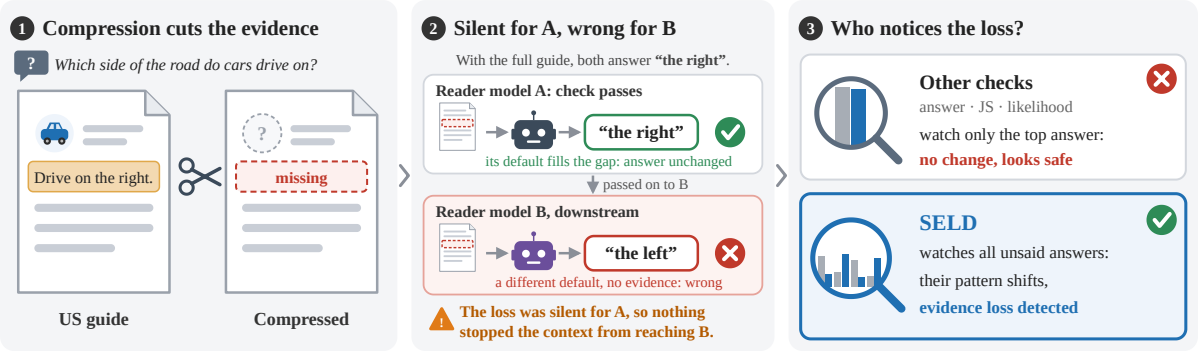


Figure 1: **Silent evidence loss and how SELD detects it** (illustrative; bars: gray for the original context, blue for the compressed one). **(1)** Compression removes the only sentence that states the answer. **(2)** With the full guide, both reader models answer *the right*. After compression, reader model A still answers *the right* from its default, so an answer check passes; reader model B, reading the same compressed context downstream, falls back on a different default and answers *the left*. The loss was silent for A, so nothing stopped the context from reaching B. **(3)** Checks that look at the top answer (answer agreement, JS divergence, answer-likelihood drop) see almost no change; in our tests they change less than when an equally long, harmless sentence is deleted. SELD compares the reader model's preferences among all the answers it leaves unsaid, which shift when the evidence is gone. A real case with measured scores is given in Section I.

ad hoc acceptance threshold when shortening code documentation (Yang et al., 2026); we instead derive the score from invariance requirements, relate it to the divergence family (Proposition 1) and evaluate it as a detector. Unlike these works, we target the case in which the answer does not move, test it with length-matched pairs, and show that much of the useful signal lies outside the tokens that probability-weighted scores emphasize.

## 3 Method

SELD has two parts: a score that compares the output of the reader model under the original and the edited context, and a paired protocol that tests whether the score tracks evidence rather than the amount of text removed. We first define the phenomenon, then the score, and finally the protocol; Figure 2 gives an overview.

**Silent evidence loss.** A frozen reader model $M$ answers a question $q$ from a context in its usual QA prompt; $\hat{a}_M(q, \cdot)$ is its complete answer after standard normalization. The original context $O$ contains evidence $E \subseteq O$ that supports the answer, and is replaced by an edited or compressed $X$.

**Definition 1** (Silent evidence loss)**.** $X$ exhibits silent evidence loss for $(M, q)$ if (i) evidence is lost, $E \not\subseteq X$, and (ii) the answer is unchanged, $\hat{a}_M(q, X) = \hat{a}_M(q, O)$, with both answers non-empty and ended by a native stop.

The definition depends on evidence annotations and answers only, not on any score. It does not require the answer to be correct, and it does not require $X$ to be unanswerable: annotated evidence marks support, not necessity. We want a score $s(X)$, computed without gold answers or evidence labels, that is higher when $X$ has lost evidence of $O$.

**The SELD score.** Before generating an answer, the reader model assigns a score to every token of its vocabulary; we treat this vector as its belief over all candidate answers, of which the generated answer is only the top entry. Removing evidence can leave the top entry in place while reshaping the rest, so a detector of silent evidence loss should (R1) read the whole belief, (R2) ignore a constant shift of all scores (the log-normalizer) and a uniform change of sharpness (the overall confidence), and (R3) compare edits of the same context on one scale. Let $\boldsymbol{\ell}_X \in \mathbb{R}^V$ denote the log-probabilities that $M$ assigns to the $V$ vocabulary tokens at the first answer position given $q$ and $X$, and let $\tilde{\boldsymbol{\ell}}_X = \boldsymbol{\ell}_X - \bar{\ell}_X \mathbf{1}$, where $\bar{\ell}_X$ is the mean of its entries. Centering removes the shift and normalization removes the scale, which gives

$$\mathrm{SELD}(X) = 1 - \frac{\langle \tilde{\boldsymbol{\ell}}_X, \tilde{\boldsymbol{\ell}}_O \rangle}{\|\tilde{\boldsymbol{\ell}}_X\|_2 \, \|\tilde{\boldsymbol{\ell}}_O\|_2}. \qquad (1)$$

SELD needs one forward pass per context and no generated answer, labels or training.

**Proposition 1.** *(i) For any* $c_1, c_2 \in \mathbb{R}$ *and* $\lambda_1, \lambda_2 > 0$*, replacing* $\boldsymbol{\ell}_X$ *by* $\lambda_1 \boldsymbol{\ell}_X + c_1 \mathbf{1}$ *and* $\boldsymbol{\ell}_O$

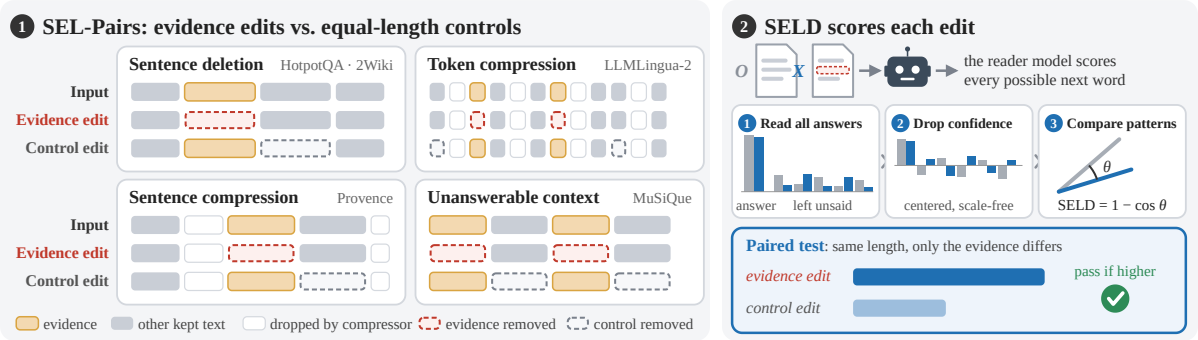


Figure 2: **SEL-Pairs and SELD** (schematic). **(1)** For each input, an *evidence edit* (red) removes evidence and a *control edit* (gray, dashed) removes other text of exactly the same token length: single-sentence deletion from the original context on HotpotQA and 2WikiMultihopQA (denoted $S$ and $C$ in the text); swaps inside LLMLingua-2 outputs at 25, 50 and 75% and inside Provence outputs ($B$ and $C$, with the output as input $A$; LLMLingua-2 swaps are refilled to equal length); and MuSiQue's official unanswerable context against a control that keeps the key paragraphs ($U$ and $T$). **(2)** SELD (step 1) reads the reader model's scores for every possible next word under $O$ and $X$, so that the answers left unsaid count; (2) centers them and ignores their scale, so that a mere change in confidence does not count; and (3) measures the angle between the two patterns. A check passes a pair when it scores the evidence edit above its equal-length control.

*by $\lambda_2 \ell_O + c_2 \mathbf{1}$ leaves SELD unchanged. Hence SELD does not depend on the softmax temperature and equals $1 - \rho(\mathbf{z}_X, \mathbf{z}_O)$, where $\mathbf{z}$ denotes the logits and $\rho$ the Pearson correlation. (ii) Let $\mathbf{a} = \tilde{\ell}_O$, $\mathbf{b} = \tilde{\ell}_X$, and let $\mathrm{JS}_T$ be the JS divergence between $\mathrm{softmax}(\mathbf{a}/T)$ and $\mathrm{softmax}(\mathbf{b}/T)$. Then $\lim_{T\to\infty} 8VT^2\,\mathrm{JS}_T = (\|\mathbf{a}\|_2 - \|\mathbf{b}\|_2)^2 + 2\|\mathbf{a}\|_2\|\mathbf{b}\|_2\,\mathrm{SELD}(X)$.*

By (ii), divergences between output distributions mix a *confidence* term (overall sharpness) with a *pattern* term, and at ordinary temperatures they weigh mostly the top few tokens; SELD isolates the pattern term (proofs in Section J). Centering is necessary: log-probabilities share a large negative offset, so the cosine of uncentered vectors is dominated by this offset and, unlike SELD, varies with the temperature.

**Paired, length-matched evaluation.** For each context we build two edits that leave the same number of reader-model tokens: $S$ removes evidence and $C$ removes the same amount of other text. A check passes a pair if it scores $S$ above $C$, with ties counted as one half. The *pass rate*, averaged over questions, is our metric; 50% is chance, and a score that only tracks how much text was removed is at chance by construction. The answer is *unchanged* for a pair when condition (ii) of Definition 1 holds for both $S$ and $C$; this group is defined from the answers alone, and answer agreement is at chance in it by definition.

## 4 Experimental Setup

The test sets below form **SEL-Pairs**. We release them as source identifiers and edit spans with scripts that rebuild every input.

**Test 1: sentence deletion.** We drew 2,414 questions from the HotpotQA (1,742) and 2WikiMultihopQA (672) training sets (Yang et al., 2018; Ho et al., 2020); none shares a paragraph with another selected question or with our development data. $S$ deletes an annotated supporting sentence and $C$ a non-supporting sentence of the same paragraph with exactly the same token length for that reader model. 1,116, 985, 1,186 and 1,116 questions qualify for the four reader models (4,459 pairs). Before running any model, we fixed the inputs, code and two main tests: SELD versus JS divergence and versus the answer-likelihood drop, on pairs with unchanged answers, averaged over reader models; eight per-reader tests count only if both main tests pass.

**Test 2: evidence swaps inside real compressed outputs.** Sentence deletion is a clean but artificial edit, so we also edit real compressor outputs. We start from the output $A$ of LLMLingua-2 (Pan et al., 2024) on the same 2,414 questions. LLMLingua-2 does not see the question, so it keeps some annotated support words and drops others. If $A$ kept $k$ support words, $B$ removes exactly these $k$ words, while $C$ removes $k$ kept non-support words whose compressor scores match them and keeps every sup-

port word of $A$. Both arms refill $k$ words that the compressor had dropped, chosen by its own scores and mostly shared, so that $A$, $B$ and $C$ have exactly the same prompt length for each reader model (Figure 2). We run this at 50% and, in a second fixed family, at 25% and 75% of the original length. For Provence (Chirkova et al., 2025), which keeps or drops whole sentences, $B$ removes one kept support sentence and $C$ one kept non-support sentence, so that the prompts of $B$ and $C$ have exactly the same length. Granite-3.3-8B-Instruct (IBM Research, 2025) serves as a fifth reader model from a fourth family at 50%. Construction reads no reader model output; it fails when no exact length match is found or word alignment is incomplete, never because of answers (Sections B and C). For each setting, the main tests (SELD versus chance, JS divergence and LUMINA-ECU on $B$ against $C$, averaged over reader models) and, where planned, per-reader tests were fixed before any reader model saw $B$ or $C$. The family for 25% and 75% also holds the three tests of an original Provence design that proved not estimable.

**Additional datasets.** MuSiQue-Full (Trivedi et al., 2022) pairs each answerable context $O$ with an officially unanswerable context $U$; a control $T$ keeps every key paragraph and matches the length of $U$ exactly (about 1,550 questions per reader model). Natural outputs of LLMLingua-2 and Provence on 4,831 questions and LongBench (Bai et al., 2024) with four compressors map the limits of SELD. Earlier HotpotQA and 2Wiki cohorts shaped the hypothesis (Section P).

**Baselines.** Besides JS divergence of the same vectors, the *answer-likelihood drop* is the mean per-token log-probability of the original answer of the reader model, given $O$ minus given $X$ (Cohen-Wang et al., 2024); *answer agreement* compares the generated answers. *LUMINA-ECU* adapts the external context utilization (ECU) score of LUMINA (Yeh et al., 2026): at the first answer position it compares the top-100 tokens under $O$ and $X$, keeping their probability mass, by a squared maximum mean discrepancy ($MMD^2$) with a shifted-cosine kernel over the input embeddings of the reader model (Section E); it is an adaptation, not a reproduction. Six *text-only scores* see only the question and the removed text: word overlap, Jaccard, query recall, BM25 (Robertson and Zaragoza, 2009), BGE cosine (Xiao et al., 2024) and the reranker bge-reranker-v2-m3 (Chen et al., 2024).

**Metrics, statistics and implementation.** All tests use the pass rate, MuSiQue with $U$ in place of $S$ and $T$ in place of $C$, and alarms across questions use the area under the ROC curve (AUROC). We resample questions within each dataset (100,000 draws) and report Bonferroni-corrected percentile intervals in brackets; family sizes are given in captions and in Section A. There was no external registration; all other analyses are post hoc and reuse saved scores, and a second implementation reproduced the reported statistics. The reader models are Qwen2.5-1.5B-Instruct, Mistral-7B-Instruct-v0.3, Olmo-3-7B-Instruct and Qwen2.5-14B-Instruct (Qwen Team, 2024; Jiang et al., 2023a; Allen Institute for AI, 2025), named below by family and size (e.g., Qwen2.5-1.5B). They run frozen in BF16 with native chat templates and no truncation, answer greedily with a 32-token cap, and we save full-vocabulary log-probabilities in FP32 at the first answer position. Code, SELPairs and the numerical supplement will be released; all appendices are in the supplementary material.

## 5 Results

### 5.1 Prevalence of Silent Evidence Loss

On the development data, deleting a key sentence and, separately, its control leaves the answer unchanged both times for 63–78% of pairs whose answers could be read. In Test 1, 66–72% of qualifying questions have at least one such pair. On MuSiQue, among questions that a reader model answers correctly with $O$ and usably with $U$, the same correct answer reappears with the unanswerable $U$ for 18.3–22.4% of questions. Unchanged answers are not a sign of safety either: where the answers to $O$, $U$ and $T$ all agree, only 15–29% are correct. When the answer of one reader model is unchanged, the same key-sentence deletion changes the answer of another model family more often than the control in 6 of 10 reader-model pairs (post hoc), although a drop in F1 for that model is established in only 1 of 10. An unchanged answer says little about the evidence.

### 5.2 Detecting Evidence Loss behind Unchanged Answers

Both main tests of Test 1 pass (Figure 3b). Among pairs with unchanged answers, SELD flags the key-sentence deletion in 82.7% of cases, JS divergence in 70.3% and the answer-likelihood drop in 53.9%,

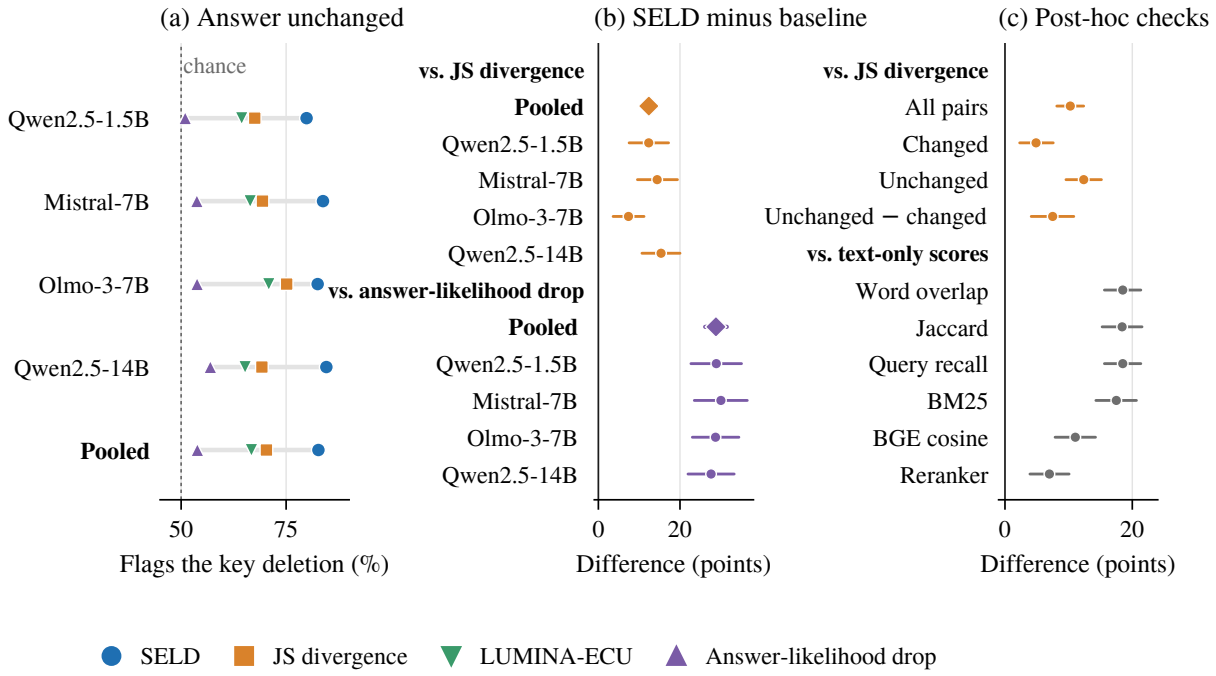


Figure 3: **Test 1: single-sentence deletion on 2,414 new questions.** (a) How often each check scores the key-sentence deletion above its equal-length control, among pairs whose answer did not change (50% is chance; LUMINA-ECU is post hoc). (b) The two main tests (pooled over reader models) and eight per-reader tests. (c) Post-hoc checks: SELD minus JS divergence by answer group and SELD minus six text-only scores among unchanged answers. Intervals are Bonferroni-corrected within each family (Section A).

close to chance (Figure 3a). SELD beats JS divergence by **12.4 points** [10.4, 14.3] and the answer-likelihood drop by **28.8 points** [26.2, 31.5], and all eight per-reader tests pass: the gain over JS divergence ranges from 7.4 (Olmo-3-7B) to 15.4 points (Qwen2.5-14B). The answer-likelihood drop scores the answer that did not change, and JS divergence weighs the same few tokens. LUMINA-ECU is weaker still, 66.7% (post hoc); SELD beats it in all eight dataset–reader cells, by 8.6 to 28.2 points among unchanged answers, with every interval above zero. LUMINA-ECU compares only the top 100 tokens, so it cannot observe a change in the pattern over the rest of the vocabulary. A real case in which SELD ranks the key deletion first and the two baselines do not is given in Section I.

### 5.3 Generalization to Real Compressors

LLMLingua-2 drops words throughout the text rather than whole sentences. At 50%, SELD ranks the evidence edit above its equal-length control in **83.2%** of cases, against 74.4% for JS divergence, 70.0% for LUMINA-ECU and 75.0% for the answer-likelihood drop (Figure 4; the last is descriptive). It beats JS divergence by **8.8 points** [7.1, 10.6] and LUMINA-ECU by **13.3 points** [11.3, 15.2], and all twelve per-reader tests pass. The removed support words matter: relative to $C$, $B$ lowers the F1 of the reader model by 25–34 points on average (descriptive). Yet for the questions whose answer stays the same under $O$, $A$, $B$ and $C$, SELD still ranks $B$ above $C$ in 77.6% of cases, against 63.9% for JS divergence and 61.7% for the answer-likelihood drop, and its lead over JS divergence is 12–15 points for every reader model, against 5–8 points when the answer changes (descriptive).

The result holds across compressors, rates and model families (Table 1). Every estimable fixed test passes: at 25% and 75% for all four reader models, for whole-sentence deletions inside Provence outputs, and for Granite-3.3-8B, a fifth reader model from a fourth family. Exact length matching is a strict requirement of the evaluation, not of SELD; it is met for 67–69% of the questions at 25% and for fewer at higher rates (Section C).

### 5.4 Ruling Out Alternative Explanations

**Text similarity and self-judgment.** In Test 1, all six text-only scores fall short of SELD among unchanged answers (Figure 3c): by 17.5–18.5 points for word-based scores and BM25, 11.1 for BGE cosine and 7.0 [3.9, 10.1] for the trained reranker.

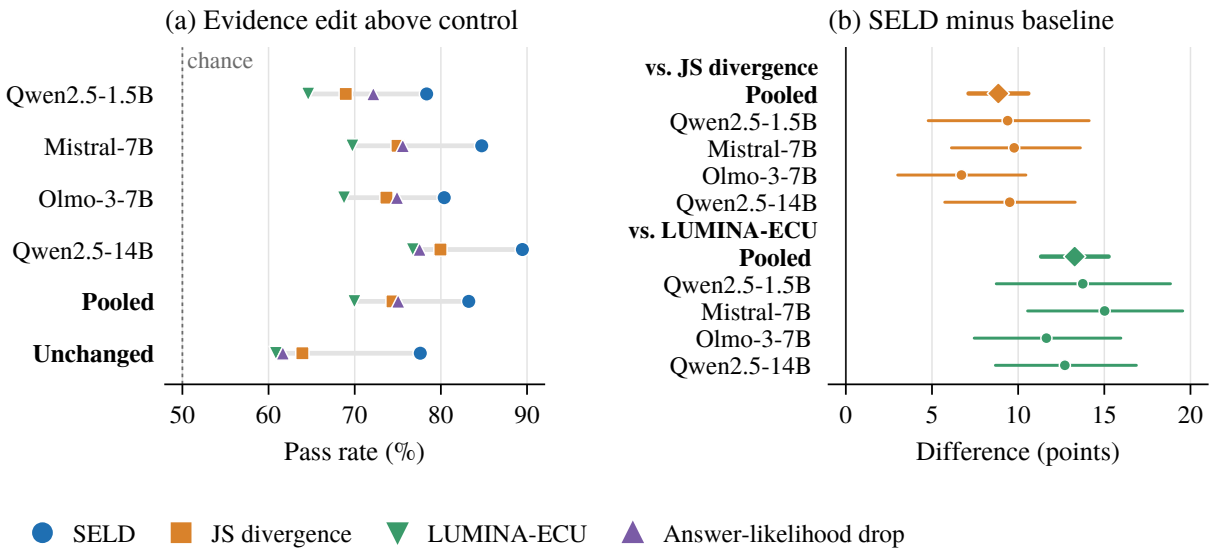


Figure 4: **Test 2 at 50%: evidence swaps inside real LLMLingua-2 outputs** (construction in Figure 2). (a) How often each score ranks the evidence edit above its control, per reader model, pooled, and pooled over questions whose answers under $O$, $A$, $B$ and $C$ are identical (descriptive). (b) Two of the three main tests (pooled) and the corresponding eight per-reader tests, with Bonferroni-corrected intervals; SELD versus chance, +33.2 [31.4, 35.0], is not shown.

| Setting | Pairs | SELD | JS | ECU | SELD − JS | SELD − ECU |
|---|---|---|---|---|---|---|
| LLMLingua-2 25% | 6,491 | **75.9** | 66.3 | 61.9 | +9.6 [7.8, 11.4] | +14.0 [12.0, 16.0] |
| LLMLingua-2 50% | 3,702 | **83.2** | 74.4 | 70.0 | +8.8 [7.1, 10.6] | +13.3 [11.3, 15.2] |
| LLMLingua-2 75% | 801 | **89.7** | 75.9 | 73.9 | +13.9 [9.5, 18.3] | +15.8 [11.1, 20.6] |
| Provence, sentences | 258 | **73.3** | 60.7 | 60.9 | +12.7 [4.2, 21.0] | +12.4 [2.2, 22.5] |
| Granite-3.3-8B, LLMLingua-2 50% | 682 | **83.1** | 71.4 | 62.3 | +11.7 [7.2, 16.3] | +20.8 [15.9, 25.8] |

Table 1: Test 2 across compressors, rates and model families: how often each score ranks the support-removing edit $B$ above the matched control $C$ (%, mean over the four reader models; Granite-3.3-8B is a single fifth reader model) and differences in points. Pairs counts question–reader pairs; each reader model has its own exactly token-matched swaps. Intervals are corrected within each fixed family: the 50%, Provence and Granite-3.3-8B settings for 3 tests each, 25% and 75% jointly for 9. Every per-reader test of the LLMLingua-2 settings passes.

In Test 2, on the questions where the control removal overlaps the question at least as much as the support removal, SELD still beats JS divergence by 6.9–13.4 points for every reader model. On the development data, SELD also beats a prompt-based sufficiency check in which the reader model judges whether the passages alone answer the question (cf. Joren et al., 2025), by 7.9–18.1 points in six of eight dataset–reader cells and ties in the other two (post hoc; Section P).

**Case selection and parametric memory.** Over all pairs of Test 1, SELD beats JS divergence by 10.3 points [8.1, 12.4]; when the answer changes, the gain shrinks to 4.9 [2.3, 7.6], and it is 7.5 points [4.1, 10.8] larger when the answer is unchanged (for three reader models, not for Olmo-3-7B). For 751 pairs with unchanged answers whose deletion removes the only mention of the answer or a bridge entity, the reader model answers correctly without any context in only 87, so closed-book recall explains few of them (post hoc). In Test 2 at 50%, qualifying questions have fewer support words on HotpotQA and shorter contexts on 2Wiki than the others, but the lead of SELD over JS divergence holds in every support-size band with at least 30 questions, by 6.0–16.0 points (descriptive, Section D).

**Beyond annotated supporting sentences.** On MuSiQue, which labels whether a context can answer at all, SELD scores the unanswerable $U$ above the length-matched $T$ more often than JS divergence and the answer-likelihood drop for every reader model, by 7.8 [6.2, 9.5] and 8.7 [6.7, 10.7] points (post hoc, Section G). For correct answers

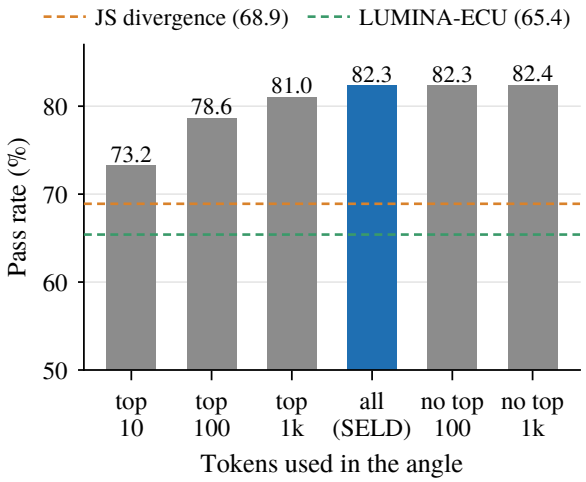


Figure 5: **The signal does not need the top tokens.** Test 1, unchanged answers: pass rate of the angle computed on subsets of the vocabulary, ranked by probability under $O$ and re-centered (unweighted mean over the eight dataset–reader cells; post hoc, descriptive).

surviving in $U$, the gain over JS divergence (+9.1 [−1.5, 19.6], 141 questions) is not established.

### 5.5 Analysis: Pattern versus Confidence

Two ingredients separate SELD from JS divergence: it compares by angle rather than by probability-weighted divergence, and it uses the whole vocabulary. Figure 5 isolates both. Computed on the top 10 tokens only, the angle already passes in 73.2% of unchanged pairs, above JS divergence (68.9%) and LUMINA-ECU (65.4%) on the same cells, so the way of comparing matters. More of the vocabulary helps further: 78.6% with the top 100, 81.0% with the top 1,000 and 82.3% with all tokens. Without the top 100 or 1,000 tokens it is unchanged (82.3% and 82.4%). What the reader model leaves unsaid carries the trace of the lost evidence. Raising the softmax temperature of JS divergence from 1 to 16 spreads its weight and improves its silent-loss pass rate in all eight development combinations, by 4.1 to 20.6 points, approaching SELD as Proposition 1 predicts, but makes it worse at detecting a *change* of answer in four of them (Section J); SELD is the temperature-free limit. The same principle carries over to the reader model's internal state: the angle between final normalized hidden states, of which the logits are a linear function, reaches 82.5% against 82.7% for SELD in Test 1 (difference +0.2 [−0.4, 0.9], post hoc). The signal thus reflects the reader model's state, not an artifact of the output layer, and SELD recovers it from the output distribution alone.

| Check | Silent loss (Test 1) | Lost answer |
|---|---|---|
| Answer agreement | 50.0 | **96.7–97.8** |
| Answer-likelihood drop | 53.9 | 81.4–83.9 |
| JS divergence | 70.3 | 81.2–83.2 |
| SELD | **82.7** | 71.8–78.6 |

Table 2: Two questions, two checks. Left: Test 1 pass rate among unchanged answers (%). Right: AUROC for flagging LongBench compressor outputs whose answer is lost (F1 drops to 0 for a question answered with F1≥50 using $O$), range over two reader models.

### 5.6 Complementarity with Answer Checks

Compression is performed by the system that holds the original context, so the reference that SELD needs is available exactly when a check is due, as in reference-based evaluation of generation. SELD and answer agreement answer two different questions. To flag outputs whose answer is lost, answer agreement is the natural check, and for silent loss it is blind (Table 2): the more a check weighs the top answer, the better it sees a lost answer and the worse it sees silent loss. The two are therefore best used together: averaging the percentile ranks of SELD and answer agreement, calibrated on one half of the development questions, beats SELD alone by 2.8 AUROC points [2.2, 3.5] and answer agreement alone by 16.2 [14.0, 18.6] on the other half, for a target positive when the answer changed or stayed despite a removed key sentence. SELD adds one forward pass per candidate (29–30 ms with Qwen2.5-1.5B; Section O). Ranking contexts of *different* questions would also need a per-question calibration (Section M).

## 6 Conclusion

In this work, we study silent evidence loss in context compression, where key evidence is removed while the answer stays the same. We introduce SELD, a training-free detector that compares a reader model's centered log-probabilities before and after compression, and SEL-Pairs, a benchmark of length-matched evidence and control edits. Across five reader models, compressors and compression rates, SELD identifies the evidence-removing edit more often than answer-, likelihood- and divergence-based checks (83% against 70% on unchanged answers). SELD needs the original context and the output distribution; future work will extend it to multi-step agents, whose compressed

histories are reused across many decisions.

## Limitations

**Evidence labels.** We use annotated supporting sentences as evidence. They mark evidence rather than strict necessity, so a flagged context is not proven insufficient, and the source of a surviving answer is analysed only in part (Section F). **Coverage.** Exact length matching inside real compressor outputs, a requirement of our evaluation rather than of SELD, covers a subset of question–reader slots, largest at moderate compression rates; within it the advantage holds across support sizes (Sections C and D). **Scope.** We study English multi-hop QA with open reader models of 1.5B to 14B parameters. SELD reads the reader model's output distribution, available for open-weight and self-hosted models, and compares each compressed context with its own original, as a paired diagnostic that complements answer checks (Section 5.6).

## Ethical Considerations

We use public research benchmarks and frozen, openly released models for offline diagnosis. No human participants were recruited and no new personal data were collected. Benchmark texts can contain names of public figures; we did not perform a content-level privacy audit. Asset licenses are listed in Section U; the numerical supplement contains scores, generated answers and opaque identifiers rather than benchmark source texts or model weights. A check that agrees with the original context does not certify factual completeness or safety and should not replace task-specific review in high-stakes settings.

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# Supplementary Material

## Guide to the Supplement

This supplement documents the evidence behind each claim of the main paper. Table S1 maps every claim to the appendices that support it and states the status of each analysis: *fixed* tests were set before the data they evaluate, *post hoc* analyses were added afterwards with their own multiple-comparison corrections, and *descriptive* results carry no inferential claim. Part I covers the main results, Part II the mechanism, Part III the scope and practical use, and Part IV development history and reproducibility. Tables and figures are numbered S1, S2, ...; references to the main paper are marked as such.

**Notation.** A *reader* is the reader model of the main paper. Qwen2.5-1.5B, Mistral-7B, Olmo-3-7B and Qwen2.5-14B denote the instruction-tuned checkpoints Qwen2.5-1.5B-Instruct, Mistral-7B-Instruct-v0.3, Olmo-3-7B-Instruct and Qwen2.5-14B-Instruct (revisions in Section U); the external judge in Section M is Qwen2.5-32B-Instruct. SELD is the score of the main paper (one minus the cosine of centered log-probabilities); JS is Jensen–Shannon divergence, Own the answer-likelihood drop, $D$ the cosine distance of final hidden states and ECU the LUMINA adaptation. *Preserved* or *silent* pairs are those whose answer did not change. Figures carried over from development runs show SELD under its working label CL.

| Claim (main paper) | Appendices | Status |
|---|---|---|
| SEL is frequent; answer checks are blind to it (Sec. 5.1) | Sections A and F | fixed, descriptive |
| SELD beats JS, answer likelihood and LUMINA on Test 1 (Sec. 5.2) | Sections A, E, F and I | fixed; ECU post hoc |
| The advantage holds at 14B scale | Section H | post hoc, fixed family |
| It carries over to real compressors (Sec. 5.3) | Sections B to D | fixed; selection descriptive |
| It is not explained by simpler factors (Sec. 5.4) | Sections F and G | post hoc |
| Mechanism and Proposition 1 (Sec. 3, 5.5) | Sections J to L | proof; post hoc |
| Complementary to answer checks (Sec. 5.6) | Sections M to O | post hoc, descriptive |
| How the hypothesis was developed | Sections P to S | fixed (dev), post hoc |
| Assets, prompts and reproducibility | Sections T and U | – |

Table S1: Where each claim of the main paper is supported.

## Part I: Main Results in Detail

### A Confirmatory Protocol and Coverage

**Inputs.** The fixed inventory has 2,414 native questions (HotpotQA train 1,742; 2WikiMultihopQA train 672). Selection used only source text: a question qualifies for a reader before any hashing, and a greedy pass over a fixed hash order then removes questions that share a paragraph title or body with an already selected question or with development data. No model output was read. Each eligible reader keeps every selected support/control reference. Controls are non-supporting sentences of the same paragraph whose deletion removes exactly as many tokens as the support deletion under the tokenizer of that reader; the fixed `relaxed_one` fallback lets the content-word overlap of the control with the question be at most one lower than that of the support sentence. Qwen2.5-1.5B and Qwen2.5-14B share a tokenizer and therefore the same references, but remain separate readers.

**Collection.** Each eligible reader produces, for $O$, every $S$ and $C$, and an empty context $E$: greedy generation with the native stop tokens and a 32-token answer cap, the full-vocabulary first-answer-position log-probabilities, the final hidden state, teacher-forced gold likelihood and the teacher-

forced likelihood of its own $O$ answer. No prompt is truncated.

**Answer layers.** Answers are normalized with the official LongBench rule (lower case, removal of ASCII punctuation and English articles, whitespace collapse) on the complete decoded response. Empty responses, responses that hit the cap and responses without a native stop are *unknown*; they are never counted as answer-changed.

**Statistics.** 100,000 bootstrap draws resample questions within each dataset, shared by all readers and comparisons. Primary intervals are Bonferroni-corrected for two tests and reader-level intervals for eight; the reader-level tests count only if both primary tests pass. A planning simulation with an 8-point effect gave a power of about 0.93 under its assumptions.

Table S2 reports the coverage of Test 1 and the absolute pass rates among unchanged answers.

| Reader | Elig. q | Refs | Pres. q | SELD | JS |
|---|---|---|---|---|---|
| Qwen2.5-1.5B | 1,116 | 1,126 | 770 | 79.87 | 67.53 |
| Mistral-7B | 985 | 997 | 666 | 83.78 | 69.37 |
| Olmo-3-7B | 1,186 | 1,210 | 786 | 82.51 | 75.13 |
| Qwen2.5-14B | 1,116 | 1,126 | 801 | 84.58 | 69.23 |
| Pooled | – | 4,459 | – | 82.69 | 70.31 |

Table S2: Coverage and absolute ranking accuracy (%) in the answer-preserving layer. The own-answer drop reaches 50.97, 53.75, 53.80, 56.96 and 53.87%. Absolute values are descriptive; the questions of different readers overlap and must not be summed.

| | SELD−JS | SELD−Own |
|---|---|---|
| Pooled (family 2) | 12.37 [10.43, 14.31] | 28.81 [26.17, 31.47] |
| Qwen2.5-1.5B (family 8) | 12.34 [7.53, 17.20] | 28.90 [22.68, 35.08] |
| Mistral-7B | 14.41 [9.57, 19.38] | 30.03 [23.51, 36.49] |
| Olmo-3-7B | 7.38 [3.61, 11.19] | 28.71 [23.03, 34.44] |
| Qwen2.5-14B | 15.36 [10.68, 20.01] | 27.62 [21.96, 33.28] |

Table S3: All ten fixed confirmatory comparisons (points, Bonferroni 95% intervals).

## B Test 2: Evidence Swaps in Compressed Outputs

**Source.** $A$ is the unmodified output of the official LLMLingua-2 compressor at a 50% rate for each of the 2,414 questions of Test 1, given the full context without the question. The word-level decisions of the compressor were replayed exactly from the saved selection mask; its continuous word scores were re-collected with the same frozen classifier, because the original run saved only the binary decisions.

**Construction.** A word belongs to the support set if all of its characters lie inside one annotated supporting sentence and to the editable non-support set if all lie inside one non-supporting sentence; titles and separators are protected. Let $K$ be the kept and $U$ the dropped words of $A$, and $k$ the number of kept support words. $B$ removes all $k$ kept support words; $C$ removes $k$ kept non-support words chosen by an ordered minimum-$L_1$ matching of compressor scores to the removed support words, and keeps every support word of $A$. Both arms then add back words from the dropped non-support pool, in decreasing compressor score, and a deterministic bounded local search (beam 4, depth 8, at most 256 complete prompts per arm) swaps single words until the complete native QA prompts of $A$, $B$ and $C$ have exactly the same number of tokens for that reader. No support word is ever added back, nothing is padded or truncated, and no reader output, score, answer or gold label is read during construction. On average the two arms share 91% of their added words (Qwen2.5-1.5B), and for Qwen2.5-1.5B $k$ has a median of 21 words.

**Coverage.** Of the 9,656 question–reader slots, 3,702 qualify (Qwen2.5-1.5B 873, Mistral-7B 839, Olmo-3-7B 1,117, Qwen2.5-14B 873). The other slots are recorded as undefined: the bounded length search was exhausted (3,638 slot-level cases), the word alignment to the source text was incomplete (2,096), a kept word crossed a support boundary (116), the original output could not be replayed (92) or there were too few non-support words (12). Construction never reads answers, so qualification does not depend on correctness. Alignment failures fall mostly on 2Wiki, where only 61–81 questions per reader qualify; no question has identical $A$, $B$ and $C$ texts for all four readers.

**Statistics.** The three main endpoints are the reader-averaged pass rate of SELD minus 50%, minus JS divergence and minus LUMINA-ECU on $B$ versus $C$, each computed on the questions where all three scores are finite. 100,000 bootstrap draws resample questions within each dataset, shared across readers; tails are $0.05/6$ for the three main tests and $0.05/24$ for the twelve per-reader tests, which count only if all three main tests pass.

Table S4 gives the per-reader results of Test 2.

| | All qualifying questions | | | | Fixed tests | |
|---|---|---|---|---|---|---|
| Reader | SELD | JS | ECU | Own | −JS | −ECU |
| Qwen2.5-1.5B | 78.4 | 69.0 | 64.6 | 72.2 | 9.4 [4.8, 14.1] | 13.7 [8.7, 18.8] |
| Mistral-7B | 84.7 | 75.0 | 69.7 | 75.6 | 9.8 [6.1, 13.6] | 15.0 [10.5, 19.6] |
| Olmo-3-7B | 80.4 | 73.7 | 68.8 | 74.9 | 6.7 [3.0, 10.4] | 11.6 [7.4, 16.0] |
| Qwen2.5-14B | 89.5 | 80.0 | 76.7 | 77.5 | 9.5 [5.7, 13.3] | 12.7 [8.7, 16.9] |
| Pooled | 83.2 | 74.4 | 70.0 | 75.0 | 8.8 [7.1, 10.6] | 13.3 [11.3, 15.2] |
| *Answers identical under O, A, B, C (descriptive)* | | | | | | |
| Qwen2.5-1.5B (250) | 71.6 | 56.4 | 54.4 | 59.6 | 15.2 | 17.2 |
| Mistral-7B (240) | 79.6 | 64.2 | 59.2 | 57.1 | 15.4 | 20.4 |
| Olmo-3-7B (288) | 76.7 | 64.6 | 61.5 | 66.3 | 12.2 | 15.3 |
| Qwen2.5-14B (326) | 82.5 | 70.6 | 68.4 | 63.7 | 12.0 | 14.1 |

Table S4: Test 2: how often each score ranks $B$ above $C$ (%), and SELD minus a baseline (points). Fixed tests: corrected for 3 (pooled) and 12 (per reader). Own is descriptive and computed where the original answer of the reader is defined.

**Further descriptive results.** *By dataset.* On HotpotQA SELD beats JS divergence by 6.0–9.8 points across readers; on the small 2Wiki sets the difference is positive for all four readers (4.5–24.2 points; 61–81 questions). *Lexical overlap.* On questions where the control removal overlaps the question at least as much as the support removal (197–260 per reader), SELD passes in 66.8, 78.2, 76.2 and 80.2% of cases against 59.9, 68.5, 68.5 and 66.8% for JS divergence. *Utility.* Relative to $C$, $B$ lowers F1 against the gold answer by 25.2, 33.0, 32.0 and 34.0 points on average (with $B$ below $C$ for 34–46% of questions) and the gold-answer log-likelihood by 1.46, 2.38, 2.36 and 3.35 nats per content token; these do not certify that every $B$ is unanswerable. *Hidden state.* The cosine distance of the final hidden state ranks $B$ above $C$ about as often as SELD (SELD minus it: $-0.5$ to $+1.5$ points over all qualifying questions, $-1.7$ to $+2.4$ among unchanged answers).

## C Test 2 at Further Rates, Provence and Granite

**Design.** The LLMLingua-2 settings at 25% and 75% reuse the construction of the 50% setting (Section B) on the outputs of the same compressor at the other rates. Their main tests (SELD versus chance, JS divergence and LUMINA-ECU, averaged over the four readers) form one family of nine together with a planned Provence setting that used the Punkt sentence splitter; that setting had 11–12 qualifying questions per reader, below the fixed minimum of 30, so its three tests are reported as not estimable and the family was not re-sized. The 24 per-reader tests of the two estimable settings form a family of 36 with the twelve not-estimable ones. The separate Provence setting uses the official sentence boundaries of the benchmark, mapped onto the tokens that Provence kept: $B$ deletes one kept supporting sentence and $C$ one kept non-supporting sentence, with exactly equal native prompt length of $B$ and $C$ per reader ($A$ is not length-matched to them). On 2Wiki every control is in the same paragraph; on HotpotQA most controls (8–11 of 13–17 per reader) come from fully non-supporting paragraphs. Qualifying questions come mostly from 2Wiki (41–57 per reader against 13–17 from HotpotQA). Granite-3.3-8B-Instruct is analysed as a single additional reader at 50% with its own family of three; the extension plan had listed Llama-3.1-8B or Gemma-2-9B as candidates, and we used the openly licensed Granite checkpoint instead. All families use 100,000 question resamples stratified by dataset; the definitions were fixed before any reader saw $B$ or $C$.

**LLMLingua-2 at 25% and 75%.** Table S5 lists the per-reader pass rates. All six main tests pass: SELD minus JS divergence is +9.6 [7.8, 11.4] at 25% and +13.9 [9.5, 18.3] at 75%, and SELD minus LUMINA-ECU is +14.0 [12.0, 16.0] and +15.8 [11.1, 20.6] (corrected for 9). All 24 per-reader tests pass (corrected for 36); for example, SELD minus JS divergence is +8.27 [4.12, 12.46] for Qwen2.5-1.5B and +10.51 [6.92, 14.13] for Mistral-7B at 25%. At 75%, only 5 2Wiki questions per reader qualify for Qwen2.5-1.5B and Mistral-7B. Where the hidden-state angle $D$ is reported (Qwen2.5-1.5B and Mistral-7B), it is within 1.7 points of SELD (descriptive).

| Rate | Reader | $N$ (coverage) | SELD | JS | ECU | $D$ |
|---|---|---|---|---|---|---|
| 25% | Qwen2.5-1.5B | 1,608 (66.6%) | 72.9 | 64.6 | 60.1 | 72.2 |
| 25% | Mistral-7B | 1,617 (67.0%) | 77.1 | 66.5 | 62.5 | 77.2 |
| 25% | Olmo-3-7B | 1,658 (68.7%) | 72.6 | 64.7 | 59.9 | – |
| 25% | Qwen2.5-14B | 1,608 (66.6%) | 81.2 | 69.3 | 65.0 | – |
| 75% | Qwen2.5-1.5B | 180 (7.5%) | 91.7 | 76.1 | 71.7 | 90.0 |
| 75% | Mistral-7B | 160 (6.6%) | 94.4 | 76.9 | 75.0 | 94.4 |
| 75% | Olmo-3-7B | 281 (11.6%) | 84.0 | 74.4 | 74.0 | – |
| 75% | Qwen2.5-14B | 180 (7.5%) | 88.9 | 76.1 | 75.0 | – |

Table S5: LLMLingua-2 at 25% and 75%: qualifying questions out of 2,414 and pass rates (%). Every SELD comparison passes its corrected test. $D$ is shown where reported in the descriptive readout.

**Provence, sentence deletion.** Qualifying questions are 66, 54, 72 and 66 for the four readers (2.2–3.0% of the questions). The four-reader means are 73.3% for SELD, 60.7% for JS divergence and 60.9% for LUMINA-ECU; all three main tests pass

(SELD minus chance +23.3 [15.6, 30.5], minus JS divergence +12.7 [4.2, 21.0], minus LUMINA-ECU +12.4 [2.2, 22.5], corrected for 3). Per reader (descriptive), SELD, JS divergence and LUMINA-ECU are tied at 65.9% for Qwen2.5-1.5B, and SELD leads for Mistral-7B (76.9 against 63.9 and 62.0), Olmo-3-7B (77.1 against 59.0 and 60.4) and Qwen2.5-14B (73.5 against 53.8 and 55.3). Among questions with unchanged answers, only 17–39 per reader remain, and for Qwen2.5-1.5B SELD is below JS divergence (55.9 against 67.6%); we therefore make no claim about silent loss for this setting.

**Granite.** 682 questions qualify (28.3%; HotpotQA 618, 2Wiki 64). SELD reaches 83.1%, JS divergence 71.4% and LUMINA-ECU 62.3%; all three tests pass (minus chance +33.1 [29.7, 36.5], minus JS divergence +11.7 [7.2, 16.3], minus LUMINA-ECU +20.8 [15.9, 25.8], corrected for 3). On HotpotQA SELD and JS divergence reach 84.5% and 71.7%; on the 64 2Wiki questions they nearly tie (70.3% and 68.8%). Among the 93 questions with unchanged answers, SELD, JS divergence and LUMINA-ECU reach 78.5%, 58.1% and 44.1% (descriptive). The hidden-state angle reaches 83.3%.

## D Test 2 Selection Check

Only 3,702 of 9,656 question–reader slots qualify at 50%. Construction reads no answers, but qualifying and non-qualifying questions differ. On HotpotQA, qualifying questions have fewer supporting sentences (standardized mean difference $-0.40$ to $-0.42$) and, where the number of kept support words $k$ is recorded, fewer support words ($-0.59$ to $-0.68$). On 2Wiki, qualifying questions have much shorter contexts ($-0.93$ to $-0.99$). We fixed five support-size bands before the analysis ($k$ = 1–16, 17–32, 33–64, 65–128, 129+). In the first three bands every reader has at least 30 questions, and SELD is above both JS divergence and LUMINA-ECU in all twelve reader–band cells, by 6.0–16.0 points over JS divergence. The two largest bands hold 1–3 questions per reader or none and are kept as such. The rank correlation between $k$ and the SELD difference between $B$ and $C$ is only 0.06–0.15, so we do not claim a dose–response relation. These are descriptive analyses; they show that the advantage holds across support sizes among qualifying questions, not that the selection is harmless.

## E LUMINA-ECU Adaptation and Results

**Definition.** LUMINA (Yeh et al., 2026) scores how much a model uses its external context by comparing the output distribution with and without the context, measuring a kernel distance between the two over token embeddings. We adapt its external-context-utilization (ECU) score to the first answer position of our setting: for $O$ and the edited context $X$ separately we take the top-100 tokens of the reader, keep their original probabilities (not renormalized), and compute the squared maximum mean discrepancy between the two weighted token sets with a shifted-cosine kernel over the input embedding matrix of the reader. Larger values mean a larger change. The definition was fixed on development data before it was applied to Test 1 and Test 2. It follows the official code where possible but is an adaptation rather than a reproduction of the full method, which also uses information about the internal knowledge of the model.

**Results.** Table S6 lists SELD minus LUMINA-ECU in the exam pass rate for every dataset–reader cell of Test 1 (post hoc). All sixteen differences for all pairs and for unchanged answers have corrected intervals above zero. On pairs whose answer changed, the differences are smaller, several intervals include zero and one point is negative (Qwen2.5-14B on 2Wiki, $-3.6$). The point estimates of JS divergence are also above LUMINA-ECU in every cell of these two groups, but by only 1.6 to 5.3 points.

| | | Test 1 (post hoc) | | Development | |
|---|---|---|---|---|---|
| Data | Reader | All | Unchanged | All | Unchanged |
| Hotpot | Qwen2.5-1.5B | 10.8 | 14.4 | 11.7 | 14.4 |
| Hotpot | Mistral-7B | 14.0 | 16.9 | 12.1 | 13.4 |
| Hotpot | Olmo-3-7B | 7.8 | 8.6 | 11.3 | 14.6 |
| Hotpot | Qwen2.5-14B | 12.2 | 14.5 | 15.9 | 20.3 |
| 2Wiki | Qwen2.5-1.5B | 15.8 | 17.5 | 13.2 | 15.4 |
| 2Wiki | Mistral-7B | 16.0 | 18.1 | 17.1 | 21.3 |
| 2Wiki | Olmo-3-7B | 15.7 | 16.7 | 11.9 | 11.3 |
| 2Wiki | Qwen2.5-14B | 25.2 | 28.2 | 17.3 | 20.6 |

Table S6: SELD minus LUMINA-ECU (points) in the exam pass rate. Every interval excludes zero (Bonferroni over the 48 comparisons of each source; 100,000 resamples).

## F Post-hoc Analyses of the Confirmatory Cohort

All analyses below reuse the saved confirmatory scores, resample the same fixed question inventory

within each dataset (100,000 draws each), were defined after the confirmatory result was known, and form their own families.

**Answer layers.** Table S7 lists the 40-comparison family (pooled and four readers, two baselines, four layers). The unknown layer is small; none of its ten intervals has a strictly positive lower limit. The direct difference between the preserved and changed layers (family of ten) is 7.49 [4.12, 10.81] for SELD−JS pooled and 9.03, 8.96, 1.34 and 10.63 points for the four readers (Olmo-3-7B crosses zero); for SELD−Own it is 23.47 [19.49, 27.42] pooled and positive for all readers. Because conditioning on unchanged answers can handicap answer-based baselines, we treat SELD−JS as the primary interpretation and SELD−Own as supporting.

| Layer | Reader | SELD−JS | SELD−Own |
|---|---|---|---|
| All | Pooled | 10.26 [8.14, 12.36] | 21.67 [18.83, 24.52] |
| | Qwen2.5-1.5B | 9.68 [5.37, 13.96] | 22.09 [16.69, 27.48] |
| | Mistral-7B | 11.98 [7.75, 16.35] | 23.00 [17.27, 28.71] |
| | Olmo-3-7B | 6.87 [3.64, 10.15] | 20.93 [16.06, 25.83] |
| | Qwen2.5-14B | 12.50 [8.27, 16.79] | 20.65 [15.59, 25.68] |
| Changed | Pooled | 4.88 [2.31, 7.58] | 5.35 [2.72, 8.08] |
| | Qwen2.5-1.5B | 3.30 [−2.50, 9.15] | 5.71 [0.00, 11.60] |
| | Mistral-7B | 5.45 [0.79, 10.96] | 6.91 [1.80, 12.78] |
| | Olmo-3-7B | 6.04 [1.96, 10.68] | 5.90 [1.45, 10.83] |
| | Qwen2.5-14B | 4.73 [−0.68, 10.50] | 2.87 [−1.89, 8.00] |
| Preserved | Pooled | 12.37 [9.58, 15.15] | 28.81 [25.07, 32.61] |

Table S7: SELD advantages by answer layer (points; post-hoc Bonferroni family of 40). Preserved-layer reader values equal the confirmatory points of Table S3 with wider intervals.

**Text-relevance baselines.** All six scores were computed for all 5,104 sentence requests with no missing values (BGE inputs at most 112 tokens, reranker pairs at most 155; no truncation). Higher means more relevant to the question. BM25 uses each own sentences of the question as the corpus ($k_1 = 1.2$, $b = 0.75$). Word count and query recall induce identical rankings within a question, so the six scores are not six independent tests.

Table S8 lists the six text-only scores against SELD.

**Lexical strata.** Strata are defined by content-word overlap of the deleted sentences with the question, without gold answers or scores. Counts of reader–dataset cells (out of 8) with positive pointwise 95% intervals, answer-preserving layer: support overlap $\leq$ control overlap, 6 (SELD−JS) and 8 (SELD−Own); equal overlap, 6 and 8; zero overlap for both sentences, 4 and 5. In the answer-changed layer the corresponding counts are much lower (2–3 of 8, and 0 in the zero-overlap stratum), and small 2Wiki cells for Mistral-7B and Qwen2.5-14B have negative points. All 424 cells of the layer and lexical analyses, including 95 without a usable interval, are in the supplement.

| Text score | $R$ (%) | SELD−score | Bonf. 6 interval |
|---|---|---|---|
| Word overlap | 64.18 | 18.50 | [15.62, 21.34] |
| Jaccard | 64.29 | 18.40 | [15.26, 21.53] |
| Query recall | 64.18 | 18.50 | [15.62, 21.34] |
| BM25 | 65.20 | 17.49 | [14.28, 20.65] |
| BGE cosine | 71.64 | 11.05 | [7.87, 14.22] |
| Reranker | 75.71 | 6.98 | [3.92, 10.07] |
| SELD | 82.69 | – | – |

Table S8: Answer-preserving layer, pooled over readers. Over all references the reranker reaches 75.40% against 85.15% for SELD. Of 24 reader-level differences, 23 have positive pointwise intervals; Qwen2.5-1.5B against the reranker is $+2.40$ [$-1.16$, 5.98].

| Dataset/reader | Exam | Answer-change | Angle−size |
|---|---|---|---|
| Hotpot/Qwen2.5-1.5B | **+7.6** | **−5.5** | **+16.3** |
| Hotpot/Mistral-7B | **+10.9** | −3.2 | **+17.3** |
| Hotpot/Olmo-3-7B | **+4.1** | **−5.4** | **+17.1** |
| Hotpot/Qwen2.5-14B | **+9.3** | −3.8 | **+16.1** |
| 2Wiki/Qwen2.5-1.5B | **+13.2** | **−18.5** | **+24.9** |
| 2Wiki/Mistral-7B | **+14.6** | −10.0 | **+18.9** |
| 2Wiki/Olmo-3-7B | **+13.2** | **−13.6** | **+15.7** |
| 2Wiki/Qwen2.5-14B | **+20.6** | +0.0 | **+25.6** |

Table S9: Spreading the weight of JS divergence (temperature 16 minus 1) on the development data: change in exam pass rate and in AUROC for detecting an answer change; last column: angle part minus size part of the score difference (points; post hoc, corrected for 40 tests; bold excludes zero).

**Temperature and geometry.** Table S9 summarizes the 40 paired contrasts. For KL and JS, raising temperature from 1 to 16 increases ranking accuracy in all eight panels. The AUROC for detecting an answer change falls significantly only for Qwen2.5-1.5B and Olmo-3-7B on both datasets; for Mistral-7B and Qwen2.5-14B the intervals cross zero, and Qwen2.5-14B on 2Wiki is $+0.02$ points.

**Automatic evidence checks.** Three automatic checks probe what the support deletions remove; none is a semantic-sufficiency label. (i) A surface rule marks a reference positive when the deletion removes the only occurrence of the answer string or of a title of the bridge entity: 1,588 positive, 1,301 negative and 1,570 undecidable

of 4,459 references (no official aliases are available). (ii) Closed-book answers: among surface-positive, answer-preserving references, only 87 are answered correctly without context, while 656 are answered incorrectly, so most preserved answers are not simply recalled. In the small subset that is surface-positive, closed-book correct and answer-preserving, SELD ranks above JS for all four readers on HotpotQA (11–39 questions each; descriptive). (iii) Renaming entities in 299 earlier questions for two Qwen2.5 readers did not establish an overall behavioural effect (difference in differences $+2.09$ and $+1.05$ points, intervals crossing zero). Finally, when the answer of one reader is preserved, the same support deletion changes the answer of another family more often than the control in 6 of 10 directed reader pairs (family of 20), but lowers the gold F1 of that reader significantly in only 1 of 10.

**Second prompt.** A reworded QA instruction for Qwen2.5-1.5B on its 311 eligible confirmatory 2Wiki questions (315 references; 945 conditions) gives $R$ of 77.33% (SELD) and 63.67% (JS) over all references, a difference of 13.67 [7.72, 19.61] (single pointwise interval, 100,000 question draws). Among the 27 answer-changed questions, SELD and JS tie and SELD is 3.70 points below the own-answer drop.

## G MuSiQue

We use the official MuSiQue-Full dev file and pair the answerable ($O$) and unanswerable ($U$) record of each of the 2,417 ids. With $D = O \setminus U$ ($k$ removed paragraphs, supporting or not) and $A = U \setminus O$ (added paragraphs), the control is $T = (O \setminus C) \cup A$, where $C$ is a set of $k$ non-supporting paragraphs of $O$. $T$ keeps every annotated supporting paragraph, has the same added paragraphs and the same number of paragraphs as $U$, and is searched (at most 128 candidates in a fixed order) for an exact match of the token length of $U$ under the tokenizer of each reader. Paragraphs are presented in a fixed hash order shared by $O$, $U$ and $T$. $R$ counts how often a score ranks $U$ above $T$. All $O \cup U$ questions form a single component of shared paragraphs, so we report points only. The MuSiQue authors report that 45 of 50 manually checked Full instances agree with the official labels; the labels are strong but not error-free.

Table S10 gives the MuSiQue differences for every reader and answer group.

| Reader | All JS | All Own | Pres. JS | Pres. Own |
|---|---|---|---|---|
| Qwen2.5-1.5B | +7.78 | +8.89 | +7.37 | +11.23 |
| Mistral-7B | +6.58 | +7.69 | +6.18 | +18.55 |
| Olmo-3-7B | +6.82 | +9.29 | +11.08 | +20.06 |
| Qwen2.5-14B | +10.15 | +8.86 | +16.31 | +17.54 |
| Macro | +7.83 | +8.68 | +10.23 | +16.84 |

Table S10: MuSiQue: SELD minus JS and minus own-answer drop (points, descriptive). Eligible questions 1,556/1,536/1,570/1,556. Answer-changed macro values are $+7.01$ and $+6.96$. The hidden-state angle $D$ is close to SELD (macro $R$ 82.97 vs. 83.21%).

Table S11 counts the correct answers that survive in the unanswerable context.

| Reader | $O$ correct, $U$ usable | Same answer in $U$ | % |
|---|---|---|---|
| Qwen2.5-1.5B | 252 | 46 | 18.3 |
| Mistral-7B | 292 | 58 | 19.9 |
| Olmo-3-7B | 286 | 64 | 22.4 |
| Qwen2.5-14B | 460 | 103 | 22.4 |

Table S11: Correct answers that survive in the officially unanswerable context. Exact match of complete normalized answers; answers that hit the cap are excluded, not counted as changed.

**Question-level intervals.** A post-hoc analysis resampled the 2,417 MuSiQue ids with replacement (100,000 draws; the same counts for all readers), and corrected 20 comparisons together: 8 per-reader and 2 mean comparisons on all qualifying questions, and the same 10 on the subset of questions answered correctly under $O$ whose complete normalized answer reappears under $U$. The subset does not require the answer to $T$ to be complete or unchanged. Questions share paragraphs, so these intervals assume exchangeable questions and are approximate.

Table S12 gives the question-level intervals for this subset.

## H 14B Scale Extension

This post-hoc extension uses Qwen2.5-14B on the full Hotpot (740 questions, 768 references) and 2Wiki (1,621 questions, 2,072 references) cohorts; inference uses the 457 and 563 questions outside the largest source component. No answer generation is added.

Each dataset uses 100,000 component bootstrap draws, and the ten fixed comparisons form one Bonferroni family. Both scale criteria pass: $D$ and SELD each exceed chance and JS. The separate SELD–KL contrasts also pass.

| Reader | $n$ | SELD / JS / Own (%) | SELD−JS | SELD−Own |
|---|---|---|---|---|
| Qwen2.5-1.5B | 46 | 50.0 / 54.4 / 52.2 | −4.3 [−28.3, 18.9] | −2.2 [−31.4, 27.0] |
| Mistral-7B | 58 | 70.7 / 63.8 / 58.6 | 6.9 [−18.3, 31.8] | 12.1 [−15.2, 38.9] |
| Olmo-3-7B | 64 | 78.1 / 60.9 / 56.3 | 17.2 [−1.5, 36.2] | 21.9 [−2.3, 45.6] |
| Qwen2.5-14B | 103 | 69.9 / 53.4 / 62.1 | 16.5 [1.0, 32.5] | 7.8 [−11.2, 26.8] |
| Mean | – | 67.2 / 58.1 / 57.3 | 9.1 [−1.5, 19.6] | 9.9 [−3.7, 23.1] |
| All, mean | – | – | 7.8 [6.2, 9.5] | 8.7 [6.7, 10.7] |

Table S12: MuSiQue, correct answers that survive in $U$ (271 reader–question observations from 141 distinct ids): rate of scoring $U$ above $T$ and paired differences (points; Bonferroni-20 intervals). Uncorrected 95% intervals for the subset means are [2.2, 15.9] and [1.1, 18.5]. The last row repeats the mean over all qualifying questions; all eight per-reader comparisons there are positive after correction.

Table S13: Complete scale-extension contrasts.

| Comparison | Difference (points) | Adjusted lower | Adjusted upper |
|---|---|---|---|
| Hotpot/D-0.5 | +42.8 | +39.2 | +45.9 |
| Hotpot/SELD-0.5 | +43.1 | +39.6 | +46.2 |
| Hotpot/D-JS | +12.1 | +7.4 | +17.1 |
| Hotpot/SELD-JS | +12.5 | +7.8 | +17.4 |
| Hotpot/SELD-KL | +11.6 | +7.1 | +16.4 |
| 2Wiki/D-0.5 | +33.0 | +28.6 | +37.2 |
| 2Wiki/SELD-0.5 | +33.0 | +28.6 | +37.1 |
| 2Wiki/D-JS | +11.2 | +5.7 | +16.6 |
| 2Wiki/SELD-JS | +11.2 | +5.9 | +16.4 |
| 2Wiki/SELD-KL | +9.8 | +4.7 | +14.9 |

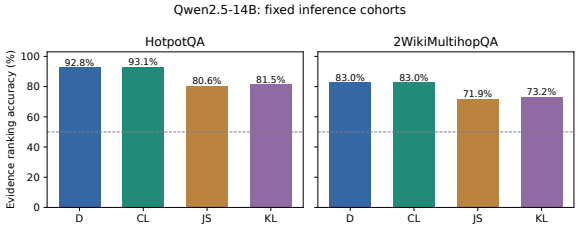


Figure S1: 14B evidence ordering on the fixed inference cohorts. These descriptive accuracies are distinct from generated-answer F1.

Figures S1 and S2 show the 14B ordering and the ten fixed comparisons.

## I Case Study

We describe a real Test 1 case for Olmo-3-7B on 2WikiMultihopQA. The question is *Were both Herbert Lang and Jason Goss, born in the same place?*; the official answer is *no*. $S$ deletes *Lang was born in Oehringen, Württemberg, Germany.* and $C$ deletes *Lang lead the expedition, until the outbreak of World War I in 1914.* from the same paragraph, with the same number of tokens. The reader answers *No* with $O$, $S$ and $C$, each a single content token ended by the native stop token. SELD is $3.35 \times 10^{-3}$ for $S$ and $0.91 \times 10^{-3}$ for $C$; JS divergence is $2.1 \times 10^{-7}$ and $9.8 \times 10^{-4}$; the answer-likelihood drop is $5.2 \times 10^{-6}$ and $3.7 \times 10^{-2}$; KL

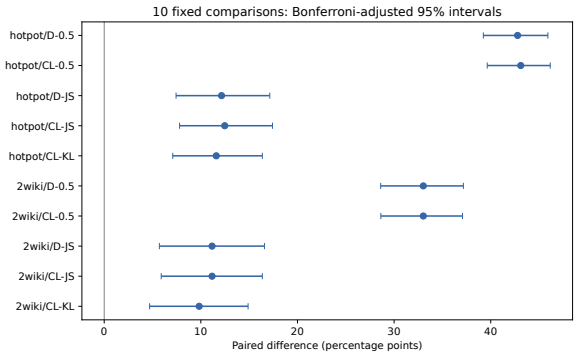


Figure S2: All ten fixed scale-extension comparisons with joint adjusted 95% intervals. No generated-answer utility claim follows from these tests.

divergence at temperature 16 and $D$ also rank $S$ first, while KL at temperature 1, the entropy change and self-judgment do not.

The case was selected for illustration from a fully enumerated pool and is not a representative sample. Among pairs with unchanged answers, 535 reader–reference pairs are ranked correctly by SELD but not by JS divergence, and 528 are ranked by SELD no higher than their control; the latter pool is the honest counterpart, and the pass rates in the main text summarize both. In Test 2, 230 question–reader slots with identical answers under $O$, $A$, $B$ and $C$ are ranked correctly by SELD but not by JS divergence, and 244 are ranked by SELD no higher than the control. For example, for *Are 11-11-11 (Film) and Bloodsport (Film) from the same country?*, Olmo-3-7B gives the same answer under all four contexts (official answer *yes*), and SELD, LUMINA-ECU, $D$ and the answer-likelihood drop rank the support-removing swap first, whereas JS divergence does not.

# Part II: Mechanism

## J Mechanism: Temperature and Proposition 1

This is a post-hoc analysis of the complete strict Hotpot deletion cohort, not a new independent confirmation. Each question retains all selected support references; references are averaged within a question, and questions have equal weight. The inference set has 457 questions, 474 references, and 391 source components. The fixed families contain 99 ranking, 75 LL, and 75 F1 contrasts, with 100,000 shared component-bootstrap draws. No new model forward or answer generation is

needed.

**Fixed finite-vocabulary expansion.** The high-temperature connection to centered logit matching is classical (Hinton et al., 2015); we use the following expansion to interpret the empirical scan, not as a new general compression guarantee.

Let $a, b \in \mathbb{R}^V$ be fixed vectors with their coordinate means removed. Set $p_T = \mathrm{softmax}(a/T)$ and $q_T = \mathrm{softmax}(b/T)$. Expanding log-sum-exp and the probabilities around the uniform distribution gives

$$\mathrm{KL}(p_T \| q_T) = \frac{\|a-b\|_2^2}{2VT^2} + O(T^{-3}), \quad (1)$$

$$\mathrm{JS}(p_T, q_T) = \frac{\|a-b\|_2^2}{8VT^2} + O(T^{-3}). \quad (2)$$

For example, $\log \sum_i e^{a_i/T} = \log V + \|a\|_2^2/(2VT^2) + O(T^{-3})$ and $p_{T,i} = V^{-1} + a_i/(VT) + O(T^{-2})$; substitution into $\sum_i p_{T,i} \log(p_{T,i}/q_{T,i})$ yields the KL coefficient. The JS coefficient follows by expanding around the mean of the two distributions. Centered logits equal centered log-probabilities exactly, because log-softmax subtracts a coordinate-constant scalar. The squared distance decomposes into $(\|a\|_2 - \|b\|_2)^2 + 2\|a\|_2\|b\|_2 \mathrm{SELD}(a,b)$, so convergence to squared distance does not generally imply convergence to cosine ranking.

**Proof of Proposition 1.** (i) Adding a constant $c$ to every coordinate of a vector leaves it unchanged after centering, and multiplying a centered vector by $\lambda > 0$ leaves its direction unchanged, so the cosine of the two centered vectors, and hence SELD, is invariant to both. A softmax temperature $T$ divides the logits by $T$, and log-softmax subtracts a coordinate-constant log-normalizer, so the centered log-probabilities at any temperature are a positive multiple of the centered logits. The cosine of two centered vectors is by definition their Pearson correlation. (ii) Multiplying the JS expansion above by $8VT^2$ gives $\|a-b\|_2^2 + O(T^{-1})$, and $\|a-b\|_2^2 = \|a\|_2^2 + \|b\|_2^2 - 2a^\top b = (\|a\|_2 - \|b\|_2)^2 + 2\|a\|_2\|b\|_2\,(1 - \cos(a,b))$. □

**The prediction confirmed.** Proposition 1(ii) predicts that tempered divergences move toward the pattern term that SELD measures as the temperature grows. The complete scan on the development deletion cohort confirms this (Table S14): JS and KL divergence rise from 78–83% at $T = 1$ to within 0.4 points of SELD at $T = 16$, and the centered squared distance matches as well. This is the theory at work rather than a competing method: tempered divergences reach SELD only after a temperature has been chosen, whereas SELD is their temperature-free limit and needs no tuning. The choice is not innocuous, because raising the temperature of JS divergence also lowers its ability to detect a changed answer in four of the eight development combinations (main paper, Section 5.5); SELD fixes the trade-off by construction. The confirmatory comparisons of the main paper use JS divergence at the standard temperature.

Only three of the 24 adjacent-temperature ranking comparisons have positive family-adjusted intervals, all for Mistral-7B. No F1 contrast among the 75 fixed comparisons has an adjusted interval excluding zero; one LL contrast (Qwen2.5-1.5B head-10 JS minus SELD) is negative. The mechanism evidence therefore concerns diagnosis, not an established general improvement in answer utility. Figure S3 plots the temperature profiles.

## K Vocabulary Ablation

For the ablation in Figure 5 of the main paper, the tokens are ranked by their probability under $O$ (ties by token id). *Top* $m$ keeps only those $m$ coordinates of both vectors; *all but top* $m$ removes them. The selected coordinates are re-centered before the angle is taken. Table S15 gives the per-cell values among unchanged answers of Test 1. The ordering top 10 $<$ top 100 $<$ top 1000 $\leq$ full holds in every cell, and removing the top 100 or top 1,000 tokens changes the pass rate by at most 1.1 points. The development data show the same pattern: the top-10 angle is below the full angle in all eight cells, and removing the top 100 or 1,000 tokens gives values close to the full angle. These are descriptive analyses; the per-cell intervals are not corrected for a new family.

## L Hidden-State Angle in Test 1

For all 4,459 pairs of Test 1, the cosine distance of the final normalized hidden state ($D$) is available. Among pairs with unchanged answers (3,057 references), the reader-averaged pass rates are 82.7% for SELD, 82.5% for $D$ and 70.3% for JS divergence. SELD minus $D$ is +0.23 points $[-0.44, 0.92]$ (post hoc, corrected for 5, 100,000 draws), and $-0.08$ when the answer changes. We therefore claim neither that SELD is better than $D$ nor that they are

| Reader | Score | $T = 1$ | $T = 2$ | $T = 4$ | $T = 8$ | $T = 16$ | SELD | Squared $L_2$ |
|---|---|---|---|---|---|---|---|---|
| Qwen2.5-1.5B | KL | 79.8 | 83.2 | 86.1 | 86.1 | 86.3 | 85.9 | 86.1 |
| Qwen2.5-1.5B | JS | 78.4 | 83.2 | 86.1 | 86.1 | 86.3 | 85.9 | 86.1 |
| Mistral-7B | KL | 83.3 | 85.6 | 90.7 | 90.9 | 91.2 | 91.2 | 91.0 |
| Mistral-7B | JS | 81.8 | 85.8 | 90.7 | 90.9 | 91.2 | 91.2 | 91.0 |
| Olmo-3-7B | KL | 82.5 | 85.4 | 89.3 | 89.9 | 89.8 | 89.8 | 89.8 |
| Olmo-3-7B | JS | 81.4 | 85.4 | 89.3 | 89.9 | 89.8 | 89.8 | 89.8 |

Table S14: Complete temperature scan on the development deletion cohort: question-weighted evidence-ranking accuracy (%). As predicted by Proposition 1(ii), tempered divergences approach SELD. Proximity is descriptive, not an equivalence test.

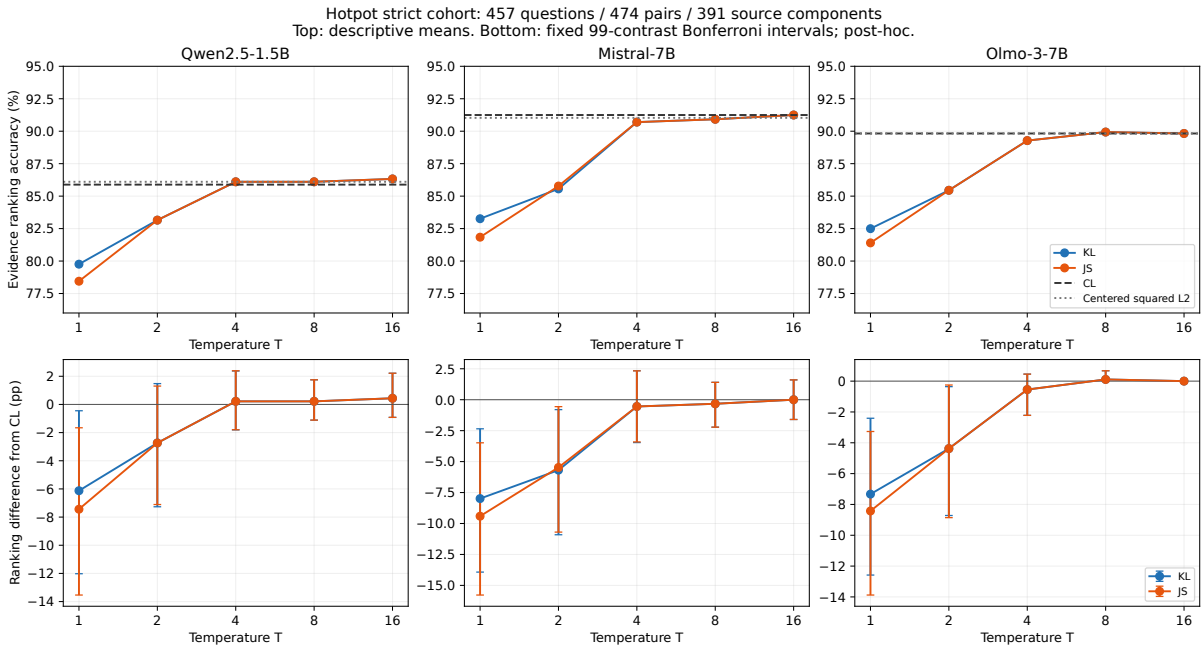


Figure S3: Temperature ranking profiles and family-adjusted contrast intervals from the complete post-hoc analysis. SELD and centered squared distance remain separate references.

| Data | Reader | top 10 | top 100 | top 1k | full | −top 100 | −top 1k |
|---|---|---|---|---|---|---|---|
| Hotpot | Qwen2.5-1.5B | 74.0 | 77.8 | 79.3 | 80.3 | 80.3 | 80.3 |
| Hotpot | Mistral-7B | 77.3 | 82.5 | 84.6 | 86.0 | 86.0 | 85.7 |
| Hotpot | Olmo-3-7B | 79.6 | 81.3 | 82.6 | 82.8 | 82.8 | 82.8 |
| Hotpot | Qwen2.5-14B | 75.8 | 82.2 | 84.1 | 85.7 | 85.7 | 85.9 |
| 2Wiki | Qwen2.5-1.5B | 71.9 | 77.6 | 78.7 | 79.1 | 79.1 | 79.1 |
| 2Wiki | Mistral-7B | 70.4 | 76.3 | 79.8 | 79.8 | 79.8 | 79.8 |
| 2Wiki | Olmo-3-7B | 73.2 | 79.4 | 80.7 | 82.1 | 82.1 | 82.1 |
| 2Wiki | Qwen2.5-14B | 63.6 | 72.2 | 78.0 | 82.6 | 82.6 | 83.6 |
| Mean | | 73.2 | 78.6 | 81.0 | 82.3 | 82.3 | 82.4 |

Table S15: Pass rate (%) of the angle on vocabulary subsets, Test 1, unchanged answers (post hoc). JS divergence and LUMINA-ECU average 68.9% and 65.4% over the same cells.

equivalent. Since the logits are a linear map of the normalized hidden state, both read the same reader state; SELD has the advantage that it needs only the output distribution.

## Part III: Scope and Practical Use

## M Natural Compressor Outputs: Scope and Practical Use

The paired design of the main paper compares two edits of the same context with exactly equal length, which isolates evidence from everything else. Natural outputs offer no such control: outputs differ in length and in what they keep, and support labels are only proxies for evidence. These analyses therefore map how SELD behaves outside the paired setting and how it is best deployed, rather than testing the paired claim.

**Data.** LLMLingua-2 at keep rates 25, 50 and 75% and Provence (official reranker, threshold 0.5) compress the full contexts of 4,831 questions (HotpotQA 1,742, 2Wiki 672, MuSiQue 2,417), giving 77,296 reader–output slots for the four readers. Provence sees the question; LLMLingua-2 does not.

**Clear-cut labels.** A fixed analysis labelled an output with an unchanged answer as keeping evidence if at least 90% of the annotated support text survived and as losing it if at most 10% survived, and required 20 outputs per class per cell. Only 6 of 48 reader–dataset–method cells qualified, all from Provence; among outputs with unchanged answers, word-level pruning almost never keeps at least 90% of the support. In all six cells SELD is above JS divergence: its AUROC is 64.1, 91.6, 66.8, 61.7, 64.5 and 93.6% against 60.6, 73.1, 58.1, 55.0, 56.4 and 64.7% for JS divergence (uncorrected pointwise intervals; the fixed pooled test is undefined). The same self-judgment prompt answered by an external judge (Qwen2.5-32B-Instruct) was also scored on these outputs: across the 23 estimable comparisons, all from Provence, SELD minus the judge is above zero in 4, below zero in 8 and undetermined in 11 (post hoc, corrected for 384). A larger external judge thus remains a strong option when its cost is acceptable and the labels are clear-cut.

**Combining with answer agreement.** On the development data we mapped SELD and answer agreement to percentile ranks calibrated on one half of the source components and evaluated on the other. The alarm target is positive when the answer changed, or when it did not change although a support sentence was deleted, and negative for an unchanged answer after a control deletion. The mean of the two percentile ranks improves the eight-cell AUROC over SELD by 2.83 [2.19, 3.52] and over answer agreement by 16.23 [13.97, 18.57] points (corrected for the 7 comparisons of this family); the maximum of the two ranks improves on answer agreement but by less than the preset 2 points on SELD. Two further candidates that subtract a closed-book reference from SELD did not improve the exam (0.54 [$-0.26$, 1.48] and $-13.2$ points).

**Graded labels across outputs (post hoc).** We then used the minimum retained fraction of the support spans as a continuous label and, within each cell, scored all pairs of outputs with unchanged answers whose overall retention differs by at most 5 points: a pair counts as concordant when the output that kept less support has the higher score. Over the 36 LLMLingua-2 cells (Provence had too few distinct labels), SELD minus JS divergence is $-0.02$ [$-0.93$, 0.88] and SELD minus LUMINA-ECU $-0.07$ [$-1.03$, 0.88] points (corrected for 2); per cell, the three concordances lie between 44.4 and 55.8%. Six sensitivity variants (mean instead of minimum retention; 2- and 10-point length windows) give intervals that all include zero. Ranking different outputs by a graded label mixes evidence loss with how much change each output tolerates, which the paired design removes; no score resolves it here.

**Choosing among outputs.** For each question and reader we picked one of the four outputs by the smallest score. Mean F1 over the four readers is 50.8 for SELD, 51.0 for JS divergence and 50.0 for LUMINA-ECU, against 43.3 for a random choice, 47.1 for always using the 50% LLMLingua-2 output and 66.4 for an oracle. The candidates differ in length, so these gains are not length-controlled.

**KV-cache compression.** We attempted SnapKV (Li et al., 2024) and KVzip (Kim et al., 2025) for all four readers. For six reader–method pairs, the implementations did not reproduce the uncompressed reader within our numerical tolerance even without compression (for SnapKV, the projections of the first layer already differed when the computation was split); Olmo-3-7B is not supported by either implementation. We therefore report no KV results.

## N Real Compressors: Complete Strong Baselines

Candidates are an abstractive prose summary, an abstractive fact list, an extract of leading source units and LLMLingua-2 at requested budgets of 256, 512, 1,024 and 2,048 tokens, for MultiFieldQA-en (150), HotpotQA (200), Qasper (200) and 2WikiMQA (200), with readers Qwen2.5-1.5B and Mistral-7B. Selection happens within each budget and is averaged over the four budgets. Scores were recomputed with full-vocabulary heads; generated answers and F1 labels are those of the original runs.

Table S16 lists all five scores on the LongBench candidates.

**Gold-answer likelihood.** A recomputation with full-vocabulary heads supports a post-hoc family of 180 contrasts. Selecting by SELD rather than JS raises gold log-likelihood by 0.196 nats/token [0.065, 0.339] for Mistral-7B; for Qwen2.5-1.5B the gain (0.049) is not established. For Mistral-7B, SELD is a better alarm than JS when the alarm target is a gold-likelihood drop of at least 1 nat/token ($+3.04$ AUROC points [0.17, 6.04]); thresholds 0.5 and 2 do not establish a difference.

| | Qwen2.5-1.5B | | Mistral-7B | |
|---|---|---|---|---|
| Score | Sel. F1 | AUROC | Sel. F1 | AUROC |
| $D$ | 32.8 | 70.9 | 37.6 | 79.3 |
| SELD | 32.9 | 71.8 | 37.4 | 78.6 |
| JS | 32.7 | 81.2 | 35.6 | 83.2 |
| Answer agreement | 34.3 | 97.8 | 37.4 | 96.7 |
| Own-answer drop | 33.2 | 81.4 | 36.1 | 83.9 |
| Random / oracle | 25.6 / 46.9 | | 26.1 / 46.8 | |

Table S16: Selected F1 and answer-loss AUROC on candidates where all five scores are valid. Selection gains over the own-answer drop: Mistral-7B SELD $+1.3$ [0.1, 2.6], Qwen2.5-1.5B SELD $-0.3$ [$-1.6$, 1.1] (pointwise).

**Scale calibration.** Dividing the angle of each candidate by the angle of a fixed zero-overlap single sentence deletion from the same question lowered answer-loss AUROC by 4.3 ($D$) and 5.0 (SELD) points for Qwen2.5-1.5B and by 8.5 and 7.8 for Mistral-7B (pointwise intervals all below zero). The calibrated angles remain 12.7–15.0 points below JS and the own-answer drop. Within a question the calibration leaves rankings unchanged.

## O Cost

We time each score on one GPU with Qwen2.5-1.5B on synthetic QA prompts of 150 and 512 tokens (median of ten runs). Per candidate, SELD takes 29–30 ms and JS divergence 35–36 ms. The answer-likelihood drop takes 23–24 ms when the original answer is cached and 84–86 ms when it must be generated, against 51–53 ms for SELD including the pass over the original context. These are synthetic measurements, not pipeline latencies.

# Part IV: Development History and Reproducibility

## P Development Cohorts

The HotpotQA dev cohort (457 questions, 474 pairs, 391 source components outside the largest component) and the 2Wiki dev cohort (563/720/456) were used during exploration. For 2Wiki we fixed a 16-test family before reading outputs (Section Q); it established angle-over-JS advantages of 9.4–12.5 points for three readers (Figure S4). With four readers and the strong baselines, the post-hoc 32-comparison family gives the values in Table S17.

| Data | Reader | SELD−JS | SELD−Own | SELD−judge |
|---|---|---|---|---|
| Hotpot | Qwen2.5-1.5B | 8.0 | 22.0 | 6.7° |
| Hotpot | Mistral-7B | 9.7 | 24.6 | 18.1 |
| Hotpot | Olmo-3-7B | 7.3 | 22.9 | 7.9 |
| Hotpot | Qwen2.5-14B | 13.5 | 22.4 | 8.9 |
| 2Wiki | Qwen2.5-1.5B | 10.0 | 21.5 | 10.8 |
| 2Wiki | Mistral-7B | 13.9 | 27.6 | 11.6 |
| 2Wiki | Olmo-3-7B | 9.4 | 27.5 | 8.2 |
| 2Wiki | Qwen2.5-14B | 15.5 | 28.4 | −0.0° |

Table S17: Development cohorts: SELD minus JS, own-answer drop and self-judgment (points, all references). Self-judgment is the reader model's log-probability of *No* minus *Yes* for the prompt *Determine whether the passages alone contain enough information to uniquely answer the question. Do not use outside knowledge or the plausibility of a guessed answer. Answer exactly Yes or No.* All intervals of the 32-comparison family exclude zero except those marked °. SELD−answer agreement is 24.6–30.5.

## Q Complete 2Wiki Confirmation

The full candidate frame has 1,621 questions and 2,072 references. Inference uses the fixed 563 questions, 720 references and 456 components outside the largest source component. The strict-length sensitivity frame contains 137 questions and 140 references outside that component; it is descriptive. The 249-question transitive-isolation cohort is another descriptive sensitivity, not a replacement inference set. Conservative length under the Qwen2.5 tokenizer does not imply the same qualification under Mistral-7B or Olmo-3-7B tokenization. No prompt or gold answer is truncated.

H1 comprises six angle-minus-JS comparisons, and passes only if at least two readers pass both. H2 tests SELD minus KL in each reader; H3 and H4 test a text-geometry score from the earliest phase of the project (persistent homology dimension of the compressed text), which we do not pursue further and report only for completeness; H5 tests KL at temperature 16 minus 1 in three readers. All 16 tests share the fixed bootstrap draws and Bonferroni 95% intervals. Qwen2.5-1.5B H4 fails and is retained. LL selection and all remaining cohort descriptions are descriptive; this set adds no generated-answer F1 or independent natural-compressor utility confirmation. Table S18 and Figure S5 list every fixed 2Wiki test.

## R Near-Length Mistral-7B Extension

This descriptive extension of Section J to a larger near-length cohort shows the same pattern: tempered divergences approach SELD, as Proposi-

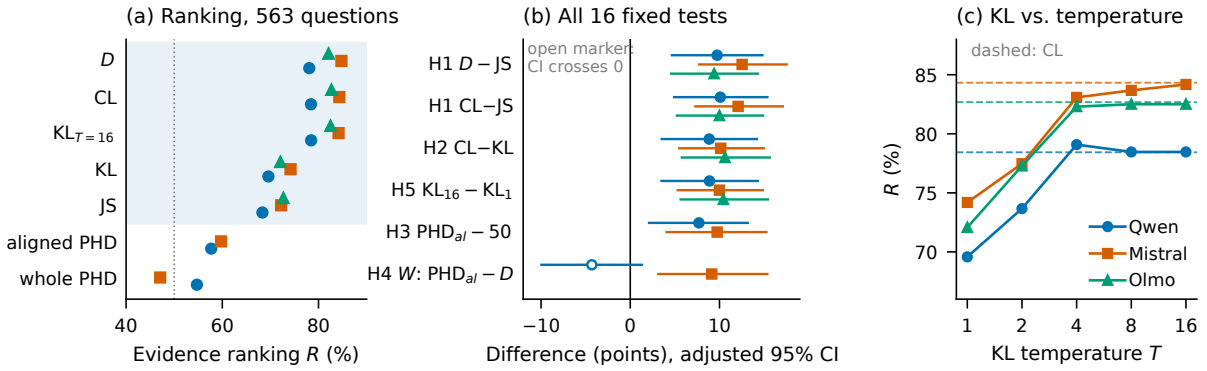


Figure S4: Earlier 2Wiki development test (563 questions, 720 pairs, 456 components). (a) Ranking accuracy; (b) all 16 fixed tests with Bonferroni 95% intervals, including the unsupported Qwen2.5-1.5B order-response test; (c) KL ranking over temperature, with SELD dashed.

Table S18: All fixed 2Wiki tests with simultaneous 95% intervals; differences are percentage points.

| Comparison | Difference (pp) | Lower | Upper |
|---|---|---|---|
| H1/Qwen2.5-1.5B/D-JS | +9.7 | +4.5 | +14.9 |
| H1/Qwen2.5-1.5B/SELD-JS | +10.1 | +4.8 | +15.5 |
| H1/Mistral-7B/D-JS | +12.5 | +7.6 | +17.7 |
| H1/Mistral-7B/SELD-JS | +12.1 | +7.2 | +17.2 |
| H1/Olmo-3-7B/D-JS | +9.4 | +4.4 | +14.4 |
| H1/Olmo-3-7B/SELD-JS | +10.0 | +5.1 | +15.0 |
| H2/Qwen2.5-1.5B/SELD-KL | +8.9 | +3.4 | +14.3 |
| H2/Mistral-7B/SELD-KL | +10.1 | +5.4 | +15.1 |
| H2/Olmo-3-7B/SELD-KL | +10.6 | +5.6 | +15.8 |
| H3/Qwen2.5-1.5B/aligned-PHD-0.5 | +7.7 | +2.0 | +13.3 |
| H3/Mistral-7B/aligned-PHD-0.5 | +9.7 | +3.9 | +15.4 |
| H4/Qwen2.5-1.5B/aligned-PHD-D | -4.3 | -10.1 | +1.4 |
| H4/Mistral-7B/aligned-PHD-D | +9.1 | +3.0 | +15.5 |
| H5/Qwen2.5-1.5B/KL16-KL1 | +8.9 | +3.4 | +14.4 |
| H5/Mistral-7B/KL16-KL1 | +10.0 | +5.2 | +15.0 |
| H5/Olmo-3-7B/KL16-KL1 | +10.4 | +5.5 | +15.6 |

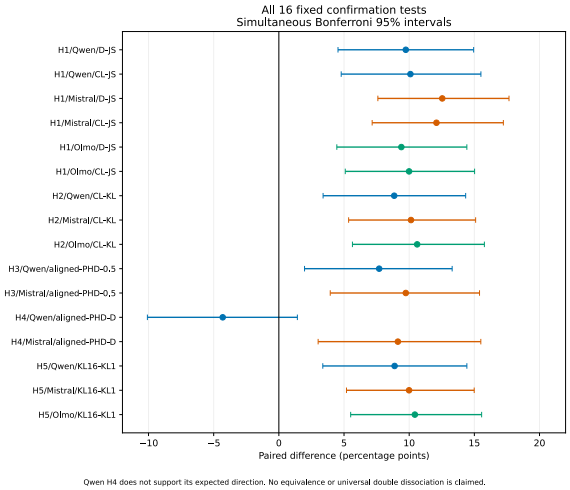


Figure S5: Every fixed 2Wiki contrast, including the unsupported Qwen2.5-1.5B H4.

Table S19: Mistral-7B near-length frame: evidence-ordering accuracy (%) of each score over all fixed pairs and the conservative and equal-length strata (descriptive).

| Score | All fixed pairs | Conservative | Equal length |
|---|---|---|---|
| JS_T1 | 80.3 | 76.3 | 79.7 |
| KL_T1 | 81.0 | 77.0 | 81.2 |
| JS_T16 | 89.3 | 83.7 | 89.7 |
| KL_T16 | 89.3 | 83.7 | 89.7 |
| SELD | 89.4 | 83.8 | 90.0 |
| D | 89.6 | 84.0 | 90.1 |
| SELD_tail1000 | 89.5 | 83.9 | 90.0 |

tion 1 predicts. It uses all 14,486 fixed same-paragraph references in 7,118 questions with saved full-vocabulary records. Outside the largest source component it retains 4,745 questions, 9,511 references and 3,064 components. No new forward, generation or inference test is performed. All 26 original scores and complete gold LL are kept. Within each length stratum references are averaged per question; strata can share questions and are not independent. The conservative positive-deletion stratum contains 2,770 questions and 3,640 references; the equal positive-deletion stratum has 362 and 372. Overall, all nine nonpositive-deletion references are retained and separately described.

Figure S6 plots these profiles.

## S Near-Length Qwen2.5-1.5B Extension

This descriptive extension repeats Section R with Qwen2.5-1.5B, using the saved full-vocabulary records of the Qwen2.5-1.5B development runs and retaining 5,552 questions and 8,224 fixed same-paragraph references. Another 6,262 references lack a support-deletion route and are reported as un-

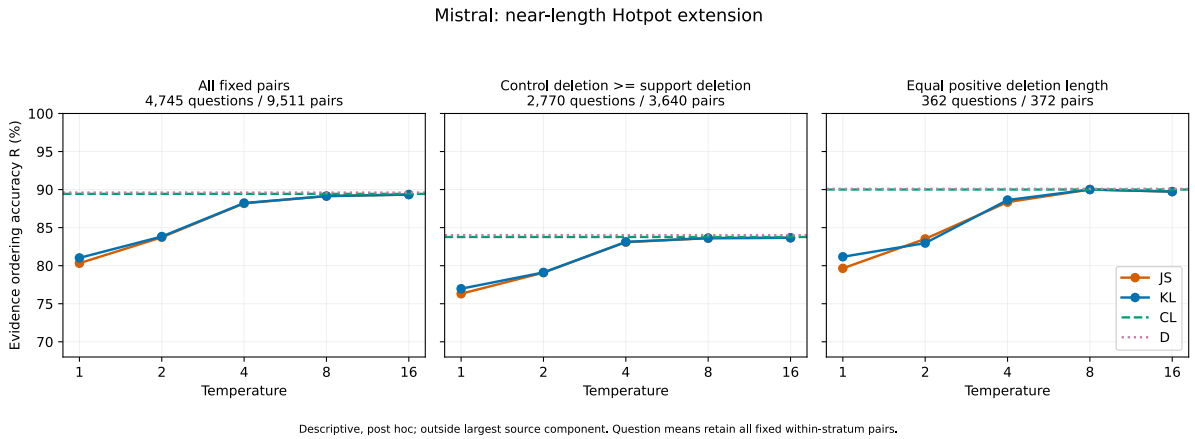


Figure S6: Mistral-7B near-length temperature profiles, entirely descriptive. The conservative stratum prevents greater support-token removal from explaining the positive evidence ordering there. This does not establish equivalence, strict monotonicity or answer-utility improvement.

Table S20: Qwen2.5-1.5B near-length frame: evidence-ordering accuracy (%) of each score over all fixed pairs and the conservative and equal-length strata (descriptive).

| Score | All fixed pairs | Conservative | Equal length |
|---|---|---|---|
| JS_T1 | 80.7 | 76.4 | 78.5 |
| KL_T1 | 80.6 | 76.5 | 79.8 |
| JS_T16 | 87.4 | 81.4 | 86.4 |
| KL_T16 | 87.3 | 81.4 | 86.4 |
| SELD | 87.4 | 81.4 | 85.9 |
| D | 87.7 | 82.2 | 87.1 |
| SELD_tail1000 | 87.4 | 81.4 | 85.9 |

covered. Outside the largest source component the frame has 3,697 questions, 5,469 references, and 2,522 components. This differs from the Mistral-7B coverage above. The conservative positive-deletion stratum has 1,778 questions and 2,072 references; exact positive matching has 458 questions and 475 references. All six nonpositive-deletion references remain in the overall frame (five outside the largest component). Within-stratum references are averaged per question, retaining all fixed pairs. No new model forward, generation or inference test is added.

The conservative JS curve rises from 76.4% at temperature 1 to 81.4% at 16, against SELD at 81.4% and D at 82.2%. It declines slightly after temperature 8, so the trend is not strictly monotone. Over all component- excluded questions, selection by SELD versus JS at temperature 1 gives mean gold LL of $-1.411628$ versus $-1.448471$ nats/token, a descriptive difference of $+0.036843$ without a new confidence interval. Neither this difference nor similar ranking values establish utility improvement or equivalence. Figure S7 plots these profiles.

## T Development Cohorts: Implementation Details

**Prompts and decoding.** Every reader uses its native tokenizer and chat template, BF16 weights and greedy decoding with a 32-token answer cap (the official caps on LongBench). First-token distributions are taken at the final prompt position, before any answer token; the gold-answer likelihood is teacher-forced over every token of each gold answer. Since $\log p_X = z_X - \mathrm{LSE}(z_X)\mathbf{1}$, centering log-probabilities equals centering logits, so SELD and JS/KL are computed from the same vectors and differ only in how they compare them.

**Deletion cohort.** Control sentences exclude every annotated supporting sentence and are chosen to match the token length of the deleted support sentence as closely as possible. The strict frame has 741 questions and 769 pairs; source components are defined by shared source titles.

**Compression candidates.** The extract keeps complete leading source units up to the budget, and LLMLingua-2 is run at the requested budget. Empty outputs and overshoots are retained.

## U Assets and Reproducibility

**Models and data.** Table S21 lists the models we use. All are used frozen, without training, fine-tuning or quantization, and only for research, consistent with their terms. HotpotQA is released under CC BY-SA 4.0, MuSiQue and Qasper under

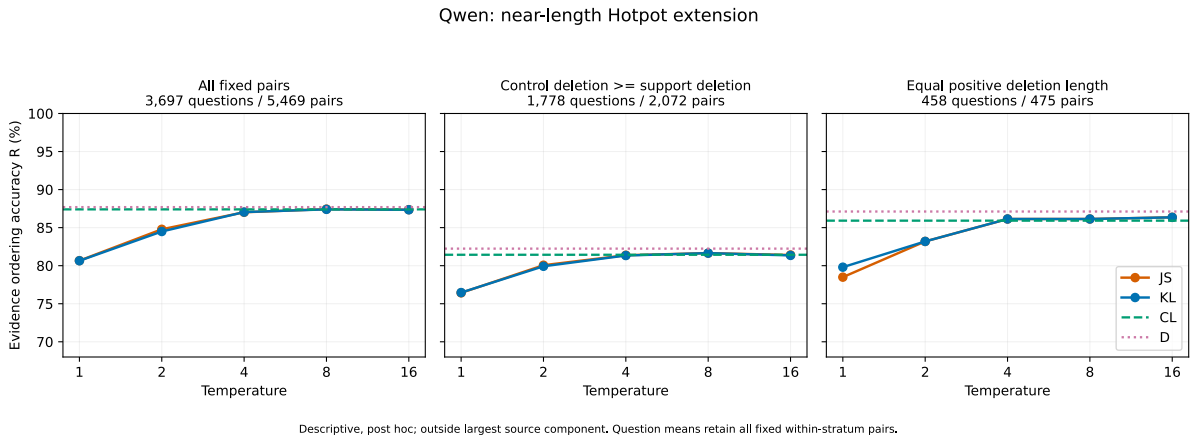


Figure S7: Full descriptive Qwen2.5-1.5B temperature profiles, including the nonmonotonic high-temperature segment and each fixed length stratum. Coverage is distinct from the Mistral-7B extension.

CC BY 4.0, and the 2WikiMultihopQA repository under Apache-2.0; the LongBench code is MIT. For 2WikiMultihopQA and LongBench the repositories do not specify a separate license for the underlying texts, and we do not redistribute them.

Table S21: Models used, all frozen; licenses as declared by the official model cards.

| Model | Use | License |
|---|---|---|
| Qwen2.5-1.5B/14B-Instruct | reader | Apache-2.0 |
| Mistral-7B-Instruct-v0.3 | reader | Apache-2.0 |
| Olmo-3-7B-Instruct | reader | Apache-2.0 |
| LLMLingua-2 (XLM-RoBERTa-large) | compressor | MIT |
| bge-base-en-v1.5 | text-only baseline | MIT |
| bge-reranker-v2-m3 | text-only baseline | Apache-2.0 |

**Computing and reproducibility.** Readers run in BF16 on RTX 3090 GPUs (24 GB); the 14B reader is split across two GPUs. All reported statistics were recomputed from the saved scores by a second, independent implementation. Saved scores, edit specifications and code will be released.

**QA prompt.** The following user message is filled and passed through each reader's native chat template.

> Answer the question based on the given passages. Only give me the answer and do not output any other words.
>
> The following are given passages.
>
> {context}
>
> Answer the question based on the given passages. Only give me the answer and do not output any other words.
>
> Question: {input}
>
> Answer: